

**Design and Implementation of an ESG Reporting Assistant
Using E-Invoice Data and Large Language Model**

BY
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A REPORT
SUBMITTED TO
Universiti Tunku Abdul Rahman
in partial fulfillment of the requirements
for the degree of
BACHELOR OF COMPUTER SCIENCE (HONOURS)
Faculty of Information and Communication Technology
(Kampar Campus)

JUNE 2026

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ACKNOWLEDGEMENTS

I would like to express my sincere thanks and appreciation to my supervisors, Prof. Ts. Liew Soung Yue who has given me the opportunity to work on a real-world company problem related to AI. This valuable experience marks an important first step in developing my knowledge and interest in the field of artificial intelligence. A million thanks to you.

Lastly, I would like to sincerely thank my parents, my family, and everyone I have met during my three wonderful years in Kampar. Their support, encouragement, kindness, and companionship have made my university journey more meaningful and unforgettable.

ABSTRACT

This project presents the design and implementation of a prototype ESG Reporting Assistant that uses e-invoice descriptions as the main input to support preliminary ESG assessment. As ESG disclosure requirements become increasingly structured, organisations require more efficient and traceable methods to interpret operational records for sustainability reporting. Although e-invoices provide standardized and verifiable transaction data, their descriptions are often brief, context-dependent, and not explicitly labelled according to Environmental, Social, or Governance dimensions. To address this issue, the proposed system integrates Retrieval Augmented Generation (RAG), a vector database, and a Large Language Model (LLM). ESG guideline documents are first collected, cleaned, segmented, embedded, and stored in a searchable knowledge base. When an e-invoice description is received, the system retrieves semantically relevant ESG reference content and combines it with the invoice information for LLM-based classification and explanation generation. The system is designed to produce ESG category recommendations, brief justifications, and summary text that can serve as a preliminary reference for ESG reporting. Preliminary work has included the construction of an ESG knowledge base and the implementation of an initial e-invoice processing workflow for retrieval, classification, and summary generation. Overall, this project aims to improve the efficiency, consistency, grounding, and traceability of interpreting e-invoice data for ESG reporting, while serving as a decision-support prototype rather than a fully autonomous enterprise reporting platform.

Area of Study: Artificial Intelligence, Natural Language Processing

Keywords: ESG Reporting, E-Invoice, Large Language Model, Retrieval-Augmented Generation, Semantic Retrieval, Vector Database, Text Classification

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LIST OF ABBREVIATIONS

<i>ESG</i>	Environmental, Social, Governance
<i>IFRS</i>	International Financial Reporting Standards
<i>LLM</i>	Large Language Model
<i>RAG</i>	Retrieval-Augmented Generation
<i>ERP</i>	Enterprise Resource Planning
<i>IRBM</i>	Inland Revenue Board of Malaysia
<i>CORD</i>	Consolidated Receipt Dataset
<i>OCR</i>	Optical character recognition
<i>NLP</i>	Natural Language Processing
<i>BERT</i>	Bidirectional Encoder Representations from Transformers
<i>NIST</i>	National Institute of Standards and Technology
<i>ESGBERT</i>	Environmental, Social, Governance Bidirectional Encoder Representations from Transformers
<i>SBERT</i>	Sentence- Bidirectional Encoder Representations from Transformers
<i>DPR</i>	Dense Passage Retrieval
<i>BM25</i>	Best Matching 25

Chapter 1

Introduction

In this chapter, the background and motivation of the project are introduced, followed by the problem statement, objectives, scope and direction, and contributions of the project. In addition, this chapter presents the organization of the report to provide a clear structure for the following chapters.

1.1 Problem Statement and Motivation

Sustainability reporting has become increasingly important as organisations are expected to disclose how environmental, social and governance (ESG) factors affect their operations, risk exposure and long-term value creation. Internationally, IFRS (International Financial Reporting Standards) S1 and IFRS S2 establish a structured basis for sustainability-related and climate-related disclosures, while Malaysia has also strengthened its sustainability reporting direction through Bursa Malaysia's reporting framework and the National Sustainability Reporting Framework. As reporting expectations become more formalised, organisations need better ways to collect, interpret and justify ESG-related information from operational records rather than relying only on manual judgement [1] [2] [3] [4].

But, during the process of organisations preparing their ESG report. They would facing the problem such as dispersed data, subjective judgment, and high manual processing costs. Especially for the e-invoice, although it will record the date of transaction, description of project, amount, supplier and validated digital records, but they are not explicitly labelled accordingly to the ESG categories. As a result, user will still need to manual labelling them whether the transaction is environmentally, socially or governance-related. This process is time consuming, inconsistent and hard to scale. Especially when the description is short, ambiguous or context-dependent [1] [2]. As a result, The core motivation of this project is how to using the e-invoice data, integrate a large language model with external knowledge retrieval mechanisms to build a system that assists in ESG materiality assessment.

Besides, although the large language model (LLM) show a strong ability in terms of text understanding, summarization, and generation. But it has a weakness as its outputs may suffer from hallucinations, insufficient grounding, or lack of traceability if it lack of external knowledge support. For ESG reporting that require credibility, explainability and evidence are critical, only relying free form model generation is insufficient. Therefore, this motivates the implementation of an ESG reporting assistant system that incorporates the concept of Retrieval-Augmented Generation (RAG). So that the model will first retrieve the relevant content from ESG guideline documents before analyzing e-Invoice descriptions, and then produce more evidence-based classifications and explanations, thereby improving the reliability and traceability of the outputs. [5] [6] [7]

1.2 Project Scope

The project focuses on the design and implementation of a prototype ESG reporting assistant that uses e-Invoice descriptions as the main system input. The system will use the vector database to store the relevant ESG guidelines, incorporate with the RAG and LLM to analyse the inputted e-Invoice description and determine whether is related to environmental, social, or governance disclosures. The final output of the system includes ESG category recommendations, brief justifications, and summary text that can serve as a preliminary report reference [2], [5].

This project is a prototype development study, focusing on validating the feasibility of integrating “e-Invoice + ESG knowledge retrieval + large language models,” rather than building a production-ready commercial platform for enterprise-wide deployment. Therefore, it does not include full ERP (Enterprise Resource Planning) integration, financial auditing, automatic regulatory report submission, handling of all enterprise document types, or fully autonomous ESG report generation. Instead, this study focuses on using e-Invoices as the primary input and leveraging selected ESG guideline documents to support decision-making and classification.

1.3 Project Objectives

The overall objective of this project is to develop an ESG reporting assistance system that integrates a retrieval mechanism with a large language model, aiming to improve the efficiency, structure, and traceability of interpreting e-Invoice data in relation to ESG relevance.

First, this project aims to construct a knowledge base built on ESG guideline documents, where relevant materials are segmented, vectorized, and stored in a searchable database. This enables the system to retrieve reference content based on semantic similarity to invoice descriptions [5]. Second, the project seeks to develop a prototype workflow in which the system can accept e-Invoice descriptions, automatically retrieve relevant ESG reference passages, and perform analysis and generation using a large language model [2], [5].

Furthermore, the system is designed to provide ESG category recommendations along with brief explanations to help users understand why a transaction is classified as Environmental, Social, Governance, or deemed irrelevant. Finally, this project emphasizes reducing the risk of unsupported or hallucinated outputs, with a strong focus on evidence grounding, explainability, and traceability rather than solely on fluency of generated text [5], [6], [7].

This project does not involve training a new foundation model, fully replacing human ESG expertise, or guaranteeing perfect automatic classification of all invoices. In other words, the focus is on developing a decision-support prototype system rather than a zero-error, fully autonomous enterprise-level ESG reporting platform.

1.4 Impact, Significance, and Contribution

The significance of this project lies in addressing the practical gap between enterprise digital transaction data and ESG reporting requirements. Although the implementation of e-Invoices has improved the standardization, verifiability, and digitalization of transaction records, companies still lack an effective method to translate raw transaction descriptions into meaningful information for ESG disclosure. By developing an ESG reporting assistance system tailored for e-Invoices, this project aims to reduce manual screening efforts, improve the consistency of preliminary assessments, and provide more structured support for the ESG reporting process [1], [2].

From an academic and technical perspective, the contribution of this project lies in applying large language models within a domain that requires strong grounding, credibility, and traceability. Compared to general chatbot-style applications, this study emphasizes how retrieval mechanisms can be used to anchor model outputs in external reference materials, ensuring that results are not merely “plausible-sounding,” but as far as possible “evidence-based.” Therefore, the contribution of this project is not only the development of a software prototype, but also the demonstration of a feasible approach that integrates e-Invoices, ESG disclosure frameworks, and Retrieval-Augmented Generation, providing a reference for future research and enterprise tool development [3] [6] [7].

1.5 Background Information

ESG is an abbreviation for Environmental, Social, and Governance, referring respectively to a company’s performance and disclosure in environmental protection, social responsibility, and corporate governance. As investors, regulators, and the public increasingly emphasize long-term sustainable development, ESG has gradually shifted from supplementary information to an essential component of corporate reporting frameworks. IFRS S1 requires entities to disclose sustainability-related risks and opportunities, while IFRS S2 further sets out specific requirements for climate-related disclosures, indicating that international sustainability reporting standards are moving toward greater consistency and decision-usefulness [3], [4].

In Malaysia, corporate sustainability disclosure is also receiving growing attention. Bursa Malaysia has issued the Sustainability Reporting Guide to assist listed companies in preparing more structured sustainability reports and to improve the quality and consistency of disclosures. With the continuous strengthening of such frameworks, companies increasingly require systematic approaches to extract ESG-relevant evidence from internal operational data, rather than relying solely on ad hoc manual judgment [1].

On the other hand, the implementation of e-Invoices has driven transaction records toward greater standardization and digitalization. According to the Inland Revenue Board of Malaysia’s e-Invoice Guideline, electronic invoices are structured, system-validated, and stored as traceable digital records. This makes e-Invoices not only useful for taxation and accounting purposes, but also a potential source of information reflecting actual business

operations. For instance, activities such as waste management, energy-efficient equipment procurement, employee training expenses, and compliance-related services may be reflected through invoice records [2].

However, the descriptions in e-Invoices are often brief, as their primary purpose is transactional recording rather than ESG disclosure. Therefore, determining whether a transaction is ESG-relevant cannot rely solely on keyword matching; it requires contextual understanding and reference to external guidelines. This is where large language models can play a role, as they are capable of understanding and generating natural language, particularly in cases where input text is ambiguous, short, or context-dependent.

Nevertheless, without external knowledge support, large language models may produce hallucinated content, insufficient justification, or outputs that are difficult to trace. Lewis et al. in their RAG study demonstrated that combining parametric models with external retrievable knowledge improves performance on knowledge-intensive tasks and enhances factual accuracy and source attribution. Similarly, the NIST Generative AI Risk Management Framework emphasizes the importance of transparency, explainability, and risk mitigation in trustworthy AI systems [5], [6]. Therefore, this study adopts Retrieval-Augmented Generation (RAG) as a foundational concept, integrating ESG reference documents into the e-Invoice analysis workflow as a core system design direction.

Overall, this project lies at the intersection of sustainability reporting, e-Invoice digitalization, and artificial intelligence-based language understanding. Rather than aiming for fully automated ESG reporting, it proposes a practical prototype that assists users in mapping e-Invoice descriptions into ESG contexts and provides more evidence-based preliminary explanations and summaries

CHAPTER 2

Literature Reviews

This literature review is organized around three key areas: first, current ESG reporting practices and e-Invoice structure; Second, the application of large language models, semantic retrieval, and Retrieval-Augmented Generation (RAG) in knowledge-intensive tasks; and third, the strengths, limitations, and research gaps in existing studies. Through this review, it becomes clearer why this study adopts an integrated approach combining “e-Invoice data + ESG guideline knowledge base + LLM,” rather than relying solely on manual review, keyword matching, or standalone language models.

2.1 ESG Reporting and E-Invoice Data

2.1.1 Development of ESG Reporting and Disclosure Requirements

In recent years, ESG (Environmental, Social, and Governance) has gradually evolved from voluntary corporate disclosure into a more structured and decision-relevant component of corporate reporting. IFRS S1 aims to require entities to disclose sustainability-related risks and opportunities in a way that is useful for general-purpose financial reporting users’ decision-making. IFRS S2 further focuses on climate-related risks and opportunities, requiring disclosures on governance, strategy, risk management, as well as metrics and targets [3], [4]. Both standards became effective on 1 January 2024, indicating that sustainability disclosure is progressively moving toward a more formal, comparable, and auditable international framework.

In Malaysia, the importance of ESG reporting has also been continuously increasing. The Sustainability Reporting Guide and related material issued by Bursa Malaysia provide listed companies with structured guidance for sustainability reporting, helping them identify material sustainability issues, organize disclosure structures, and improve report quality [1], [7]. In addition, Malaysia’s National Sustainability Reporting Framework (NSRF) emphasizes that companies should provide more consistent, comparable, and reliable sustainability information in order to enhance market transparency and investment attractiveness [17]. This demonstrates

that, from both international and local perspectives, companies increasingly require more effective methods to translate internal operational data into information that supports ESG disclosure.

2.1.2 E-Invoice Structure and Pattern

In this context, e-Invoice emerges as a promising data source. According to the e-Invoice Guideline ver. 4.6 issued by the Inland Revenue Board of Malaysia (IRBM), an e-Invoice is a system-validated electronic record of transactions that can be generated, transmitted, and managed in a structured manner, and integrated with systems via the MyInvois platform or API [2]. Compared to traditional paper-based invoices or unstructured documents, e-Invoices are characterized by their digital, standardized, verifiable, and traceable nature. They typically contain key data elements such as transaction date, buyer and seller information, item descriptions, amounts, and tax-related details. (Figure 2.1.1 E-Invoice)

FOR ILLUSTRATION PURPOSES ONLY

Food Eatery Sdn Bhd
 1, Tingkat 2, Jin Jaya, 40150 Selangor, Malaysia
 0312345678 (Office) / 6012334567890 (Mobile)
 Food.eatery@gmail.com

Supplier TIN: E100000000030
 Supplier Name: ABC Advisory Ltd
 Supplier Registration Number: NA
 Supplier SST ID: NA
 Supplier Business Address: 1, Street Avenue, NOP 123, England
 Supplier Contact Number (Office): 441234567890
 Supplier E-mail: ABC.advisory@gmail.com
 Supplier MSIC code: 00000
 Supplier business activity description: NA

Buyer TIN: C987654321120
 Buyer Business Registration Number: 29802101000023
 Buyer SST ID: L10-5621-78000000

E-INVOICE
 e-Invoice Type: 11 – Self-billed Invoice
 e-Invoice version: 1.0
 e-Invoice code: INV00005
 Unique Identifier No: 123456789-2023-7654231
 Original Invoice Ref. No.: Not Applicable
 Invoice Date and Time: 2025-08-01 13:14:52

Classification	Description	Quantity	Unit Price	Amount	Disc	Tax Amount	Total Product / Service Price (incl. tax)
035	Provision of legal advisory services	1	RM200,000.00	RM200,000.00	-	RM16,000.00	RM216,000.00
	Subtotal			RM200,000.00	-	RM16,000.00	RM216,000.00
	Total excluding tax						RM200,000.00
	Tax amount (SST)						RM16,000.00
	Total including tax						RM216,000.00
	Total payable amount						RM216,000.00

Total Product / Service Price	Tax code	Tax type	Tax Rate	Tax amount
RM200,000	02	Service Tax	8.00%	RM16,000

Digital Signature:
 7e83e05cce9b5db17ac0deec3b7c5gyba983f6dc50531c7a919f28d5fb369ev4
 Date and Time of Validation: 2025-08-01 13:14:53
 This document is a visual presentation of the e-Invoice



Figure 2.1.1 E-Invoice

From a research perspective, the value of e-Invoices extends beyond tax compliance purposes; they also serve as an important data source that closely reflects day-to-day business operations. Compared with annual reports, sustainability reports, or news articles, e-Invoices are more directly tied to actual transactional events, and therefore may contain early indicators relevant

to ESG considerations. For example, activities such as energy-efficient equipment procurement, waste management services, employee training expenditures, occupational health and safety supplies, and compliance-related services may be reflected through invoice item descriptions [1], [2]. Hence, if these transaction descriptions can be systematically interpreted, e-Invoices have the potential to serve as a valuable input for preliminary screening and supporting analysis in ESG reporting processes.

2.1.3 Advantages and Limitations of Document Intelligence and Invoice Research

Research on invoices and receipts has demonstrated significant strengths in the areas of business document structure understanding, field extraction, and multi-layout robustness. Benchmarks such as DocILE provide large-scale datasets for business document processing, while systems such as CORD integrate OCR with semantic parsing, both of which offer substantial contributions to the automated processing of commercial documents [13], [14]. If the objective of this study were limited to extracting invoice numbers, monetary values, or tabular fields from images, these works would serve as essential references.

However, their limitation lies in the fact that invoices are primarily treated as structured information extraction targets rather than as carriers of semantic or disclosure-related meaning. Even if a system can accurately extract elements such as vendor, date, and total amount, this does not necessarily imply that it can determine whether a transaction is relevant to ESG reporting. In other words, invoice extraction focuses on “what is contained in the document,” whereas this study is concerned with “what these contents mean in an ESG context.” This fundamental difference clearly distinguishes the present study from prior research on invoice processing.

2.2 Applications of NLP and Large Language Models in ESG Text Understanding

2.2.1 Traditional Way: Manual Judgement and Keyword

In many organizations, ESG data identification still largely relies on manual judgment, keyword lists, Excel-based filtering, or static rule systems. These approaches are advantageous

in that they are simple to implement, easy to interpret, and lower cost, making them suitable for small-scale datasets. However, their limitations are also evident: when textual expressions are diverse, context-dependent, rich in synonyms, or very short, keyword-based methods are prone to both false negatives and false positives. In addition, rule-based systems often lack flexibility, and their maintenance cost can increase significantly when new terminology, abbreviations, or industry-specific expressions are introduced [1], [2].

For short texts such as e-Invoice descriptions, keyword-based approaches are particularly vulnerable to ambiguity at the boundary level. For instance, terms like “waste services” may clearly indicate environmental relevance, whereas descriptions such as “facility maintenance,” “compliance support,” or “training package” may or may not be ESG-related depending on the broader business context and disclosure requirements. In other words, while traditional methods can perform well in simple and explicit cases, they are generally less robust when dealing with semantically implicit transactional descriptions.

2.2.2 Prior Research on NLP Models in the ESG Domain

With the increasing demand for ESG text analysis, a growing number of studies have adopted NLP models to address ESG-related classification tasks. The ESGBERT model proposed by Mehra, Louka, and Zhang extends the BERT architecture (Figure 2.2.1 BERT Architecture) by further training it on ESG-domain-specific corpora. Their findings indicate that such domain-adapted pretraining approaches can outperform the original BERT model and other baseline methods in ESG classification tasks [11]. This suggests that when the target task exhibits strong domain-specific linguistic characteristics, leveraging models trained on in-domain corpora can effectively improve classification performance.

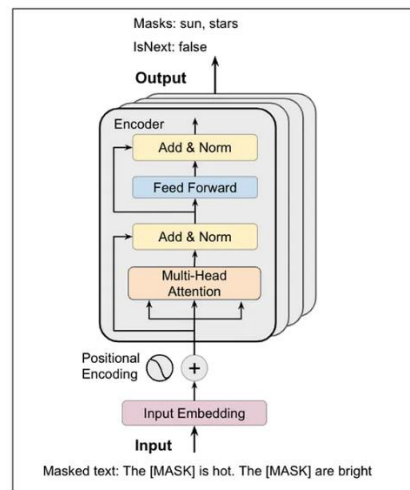


Figure 2.2.1 BERT Architecture

Another large-scale research direction involves using extensive collections of corporate reports and news articles to develop ESG sub-domain models. Schimanski et al. trained models on more than 13.8 million texts, constructing separate representations for Environmental, Social, and Governance domains, and building corresponding classification datasets to quantify ESG-related communication in corporate disclosures [12]. The strength of this line of work lies in its large-scale data coverage and its ability to demonstrate the applicability of NLP methods in ESG communication measurement.

However, a common characteristic of these prior studies is that they primarily focus on relatively long and well-structured textual sources, such as annual reports, news articles, regulatory filings, and press releases [11], [12]. In contrast, this study deals with much shorter and more fragmented e-Invoice descriptions. The challenges in long-text classification and short-text semantic interpretation are not identical; the latter often lacks sufficient contextual information and is more dependent on external knowledge to infer meaning. Therefore, although existing ESG NLP literature provides valuable methodological foundations, it cannot be directly considered a complete solution for the e-Invoice-based ESG classification setting.

2.2.3 Potential and Limitations of Large Language Model

Large Language Models (LLMs) demonstrate strong generalization capabilities in tasks such as summarization, classification, explanation generation, and question answering, making them highly suitable as a core component for interpreting the semantic meaning of invoice

descriptions in this study. Compared to static keyword-based approaches or traditional rule systems, LLMs are able to capture more subtle semantic variations with minimal handcrafted rules and generate coherent natural language explanations. In the context of an ESG assistance system, this means that the model can not only provide category predictions but also articulate why a given transaction may be relevant to Environmental, Social, or Governance dimensions.

However, prior studies and official risk frameworks also highlight the limitations and risks of using LLMs in isolation. Lewis et al. note that although parametric language models store substantial knowledge within their parameters, their ability to access and precisely manipulate knowledge in knowledge-intensive tasks remains constrained. Similarly, the NIST Generative AI Profile emphasizes that generative AI systems require careful consideration of transparency, traceability, reliability, and risk management [5], [6]. Therefore, if a system relies solely on an LLM to perform ESG classification on e-Invoice data without external references, it may produce insufficiently grounded reasoning, overgeneralized inferences, or outputs that are difficult to audit.

2.2.4 Advantages and Limitations of ESG Reporting and ESG NLP Research

The main strength of existing ESG text analysis research lies in demonstrating that NLP methods can indeed support sustainability-related text tasks that were traditionally highly dependent on manual effort. Whether through domain-specific language models such as ESGBERT or large-scale ESG sub-domain models pretrained on corporate reports and news articles, prior studies have shown that automated ESG text identification is feasible [11], [12]. These works provide important insights for this study, namely that ESG-related text is not inherently uncomputable; given sufficient domain-specific corpora, models are capable of learning meaningful semantic patterns.

However, several limitations remain evident. First, most studies focus on relatively long and well-structured text sources such as corporate reports, news articles, or disclosure documents, rather than short transactional descriptions. Second, many studies emphasize classification accuracy or quantitative evaluation rather than providing traceable explanations at the individual transaction level. Third, these approaches are often not directly grounded in formal ESG guidelines during inference, leaving room for improvement in explainability, particularly

in terms of “why a certain classification is made” [11], [12]. In contrast, this study places greater emphasis on short-text scenarios, evidence-based reasoning, and transaction-level explanation.

2.3 Retrieval-Augmented Generation (RAG) and Semantic Retrieval

2.3.1 Basic Concept of RAG

Retrieval-Augmented Generation (RAG) is based on the idea of first retrieving relevant information from an external knowledge base before generating an answer, and then feeding both the retrieved content and the user query into a generative model for processing. Lewis et al. in the original RAG study point out that this approach can provide external knowledge and source grounding for knowledge-intensive tasks, thereby improving the factual accuracy and specificity of generated outputs [5]. This is particularly important for the present study, as ESG relevance assessment is not merely a language fluency task, but one that requires “evidence-based explanation.”

Subsequent review studies further highlight that RAG has gained significant attention because it helps mitigate several typical limitations of large language models, including outdated knowledge, hallucination, lack of transparency in reasoning, and difficulty in traceability [10]. In high knowledge-density domains such as regulations, disclosure standards, and organizational internal documents, RAG is generally more suitable than relying solely on parametric knowledge within the model. Since this project aims to establish semantic connections between e-Invoice descriptions and ESG guideline content, adopting RAG as the core approach is theoretically well justified [5], [6], [10].

2.3.2 Dense Retrieval, Semantic Embeddings, and Vector Search

A fundamental component of RAG systems is the representation of text as embeddings that enable similarity computation. Sentence-BERT (SBERT), proposed by Reimers and Gurevych, generates sentence embeddings using a Siamese BERT architecture, significantly reducing the computational cost of sentence similarity comparison compared to traditional BERT, while

improving the feasibility of semantic retrieval [8]. Such sentence embedding techniques provide an important foundation for this study, as matching between e-Invoice descriptions and ESG guideline passages is often based on conceptual similarity rather than exact lexical overlap. (Figure 2.3.1 BERT and SBERT architecture)

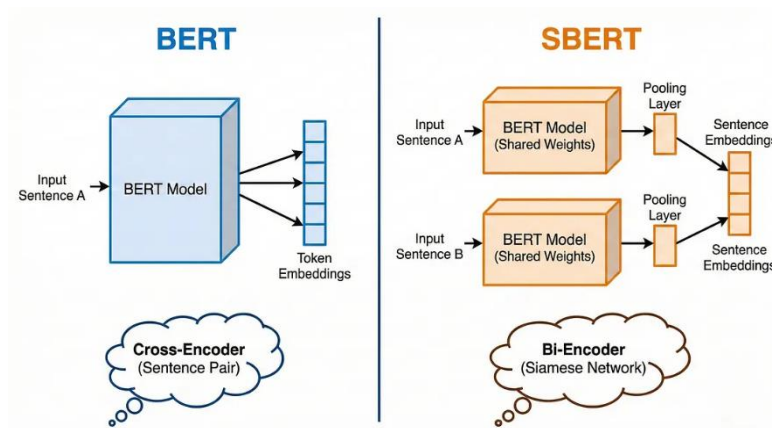


Figure 2.3.1 BERT and SBERT Architecture

In terms of retrieval mechanisms, Dense Passage Retrieval (DPR) (Figure 2.3.2 Dense Passage Retrieval (DPR)) proposed by Karpukhin et al. demonstrates the effectiveness of dense representations in open-domain question answering tasks, showing that dense retrievers significantly outperform traditional sparse methods such as BM25 in passage retrieval accuracy [9]. This is particularly relevant to this study: keyword-based matching may fail to retrieve ESG-relevant content when expressions differ lexically but are semantically related, such as “training expense,” “safety kit,” or “waste handling service,” whereas semantic retrieval is more capable of capturing such conceptual relationships.

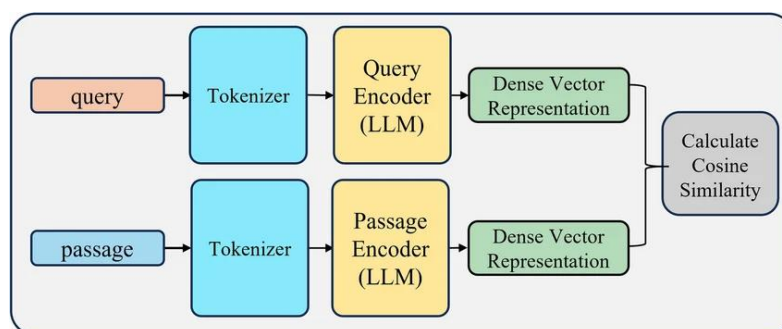


Figure 2.3.2 Dense Passage Retrieval (DPR)

From a broader perspective of RAG research, the integration of semantic embeddings, vector indexing, and external knowledge bases has become a key paradigm in modern knowledge-augmented systems [10]. Therefore, the use of a vector database to store ESG guideline

documents and perform retrieval based on semantic similarity is not only an engineering choice but also consistent with current developments in RAG research.

2.3.3 Importance of Document Chunking Strategies

In addition to embedding models and retrieval methods, the performance of RAG systems is highly dependent on how documents are segmented. Hearst's TextTiling method demonstrated that text can be segmented into topically coherent units based on subtopic boundaries, producing segments that better reflect human perception of discourse structure [15]. This is particularly important for ESG guideline documents, which often contain definitions, principles, case studies, explanations, and tables within the same document; improper chunking may lead to the mixing of unrelated concepts within a single retrieval unit, thereby reducing retrieval quality.

More recent empirical studies further show that chunking is not necessarily "the more complex, the better." Qu, Tu, and Bao conducted a systematic comparison of different chunking strategies in RAG systems and found that semantic chunking may offer advantages in documents with high topical diversity, but such benefits are highly context-dependent and do not always outweigh its higher computational cost. In more realistic, non-synthetic corpora, fixed-size chunking may in some cases perform more consistently [16].

This suggests that in designing the ESG guideline knowledge base for this study, it is inappropriate to assume that semantic chunking is always superior. Instead, a trade-off must be made based on document structure, retrieval objectives, and system cost. For this study, chunking is not merely a technical optimization issue, but a factor that directly affects whether the system can retrieve ESG reference passages that are sufficiently complete yet not overly verbose. If chunks are too large, multiple ESG subtopics may be mixed, increasing retrieval noise; if they are too small, necessary contextual information for explanation may be lost. Therefore, balancing structural integrity, semantic coherence, and retrieval efficiency is a key design consideration in this research [10], [15], [16].

2.3.4 Advantages and Limitations of RAG Methods

The primary advantage of Retrieval-Augmented Generation (RAG) lies in its ability to provide external knowledge and traceable evidence to generative models, making it particularly suitable for knowledge-intensive tasks. Studies by Lewis et al. and subsequent reviews indicate that RAG can improve the factual accuracy of generated outputs while mitigating issues such as static parametric knowledge, difficulty in updating information, and hallucination [5], [10]. In the context of this study, this implies that the system does not need to rely on intuitive or implicit reasoning to determine whether an invoice is ESG-relevant; instead, it can first retrieve the most relevant ESG guideline passages and then generate explanations based on them.

However, RAG is not without limitations. Its performance is highly dependent on retrieval quality, which is influenced by factors such as embedding models, indexing strategies, chunk size, overlap configuration, and document structure [8], [9] [10], [15], [16]. If the knowledge base is poorly segmented, the system may retrieve passages that appear relevant but are actually contextually misaligned. Similarly, inappropriate embedding or retrieval configurations may fail to retrieve truly important ESG references. Therefore, although this study adopts RAG as a core approach, it also places strong emphasis on document preprocessing and retrieval design, rather than treating RAG as a universally reliable or fully automatic solution.

2.4 Summary

Overall, prior research has made progress in three main areas: ESG text classification, invoice document processing, and RAG-based system design. However, a gap still exists between these areas. ESG NLP studies mainly focus on long disclosure texts, invoice research emphasizes structured information extraction, and RAG research is often designed as a general framework or for question-answering tasks.

Relatively few studies have combined e-Invoices as the primary input, ESG guideline documents as the retrieval knowledge base, and large language models to generate ESG classifications with evidence-based explanations. This gap defines the focus and innovation of the present study [2], [5], [10], [11], [12], [14].

Chapter 3

SYSTEM METHODOLOGY/APPROACH

3.1 System Design Diagram

3.1.1 System Architecture Diagram

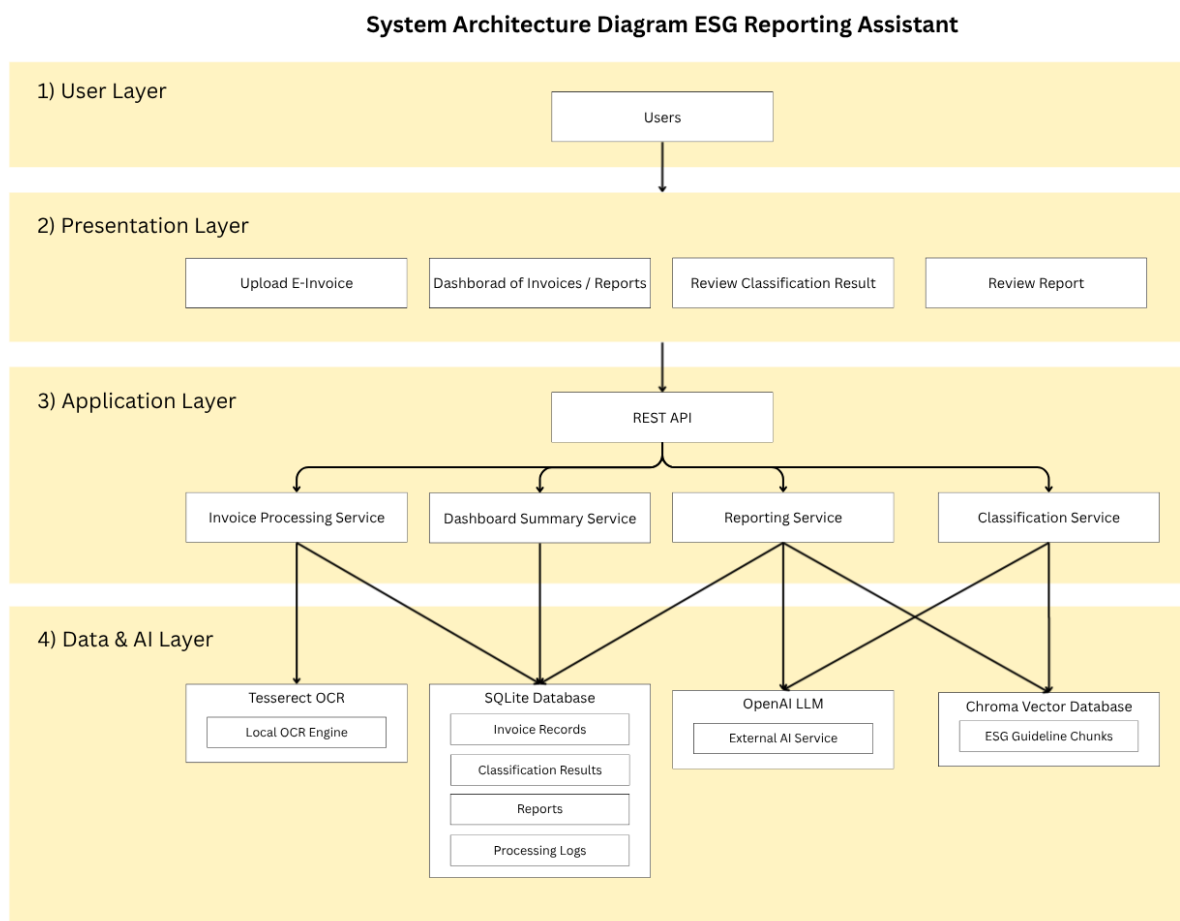


Figure 3.1.1 System Architecture Diagram

Figure 3.1.1 illustrates the overall system architecture of the ESG Reporting Assistant. The system adopts a layered architecture consisting of four main layers: the User Layer, Presentation Layer, Application Layer, and Data & AI Layer. This separation allows the user interface, backend processing logic, data storage, and AI-related components to operate

independently while communicating through clearly defined interfaces, thereby improving the maintainability, scalability, and modularity of the system.

The **User Layer** represents the end users who interact with the ESG Reporting Assistant. Users access the system through the web-based interface to perform the main functions provided by the system. These functions include uploading e-invoices, monitoring invoice and report information, reviewing ESG classification results, and reviewing generated ESG reports. The User Layer serves as the entry point to the system and communicates with the functions provided through the Presentation Layer.

The **Presentation Layer** was developed using Next.js and provides the graphical user interface for interacting with the system. It consists of the Upload E-Invoice interface, Dashboard of Invoices and Reports, Review Classification Result interface, and Review Report interface. The presentation layer is responsible for displaying information to users and collecting user inputs without directly performing the underlying processing logic. User requests and data are transferred to the backend through REST API requests, allowing the frontend to remain separated from the application processing components.

The **Application Layer** was implemented using FastAPI and contains the main backend processing logic of the system. The REST API acts as the communication interface between the Presentation Layer and the backend services. The Invoice Processing Service processes uploaded e-invoices, performs text extraction through Tesseract OCR when required, and stores the processed invoice information in the database. The Dashboard Summary Service retrieves stored information to provide summary data for the dashboard. The Classification Service performs ESG classification by retrieving relevant ESG guideline information from ChromaDB and using the OpenAI LLM to generate classification results and explanations. The Reporting Service uses stored invoice and classification information together with relevant ESG guideline content and the OpenAI LLM to generate structured ESG-related reports.

The **Data & AI Layer** provides the data storage, knowledge retrieval, optical character recognition, and AI capabilities required by the system. SQLite is used as the relational database to store invoice records, classification results, generated reports, and processing logs. Tesseract OCR functions as a local OCR engine for extracting textual information from uploaded invoice documents when necessary. ChromaDB serves as the vector database for

storing ESG guideline chunks and enables the Classification and Reporting Services to retrieve relevant ESG knowledge during processing. The OpenAI LLM functions as an external AI service that supports ESG classification, explanation generation, and report generation. Together, these components provide the data persistence and AI capabilities required to support the complete workflow of the ESG Reporting Assistant.

3.1.2 Use Case Diagram and Description

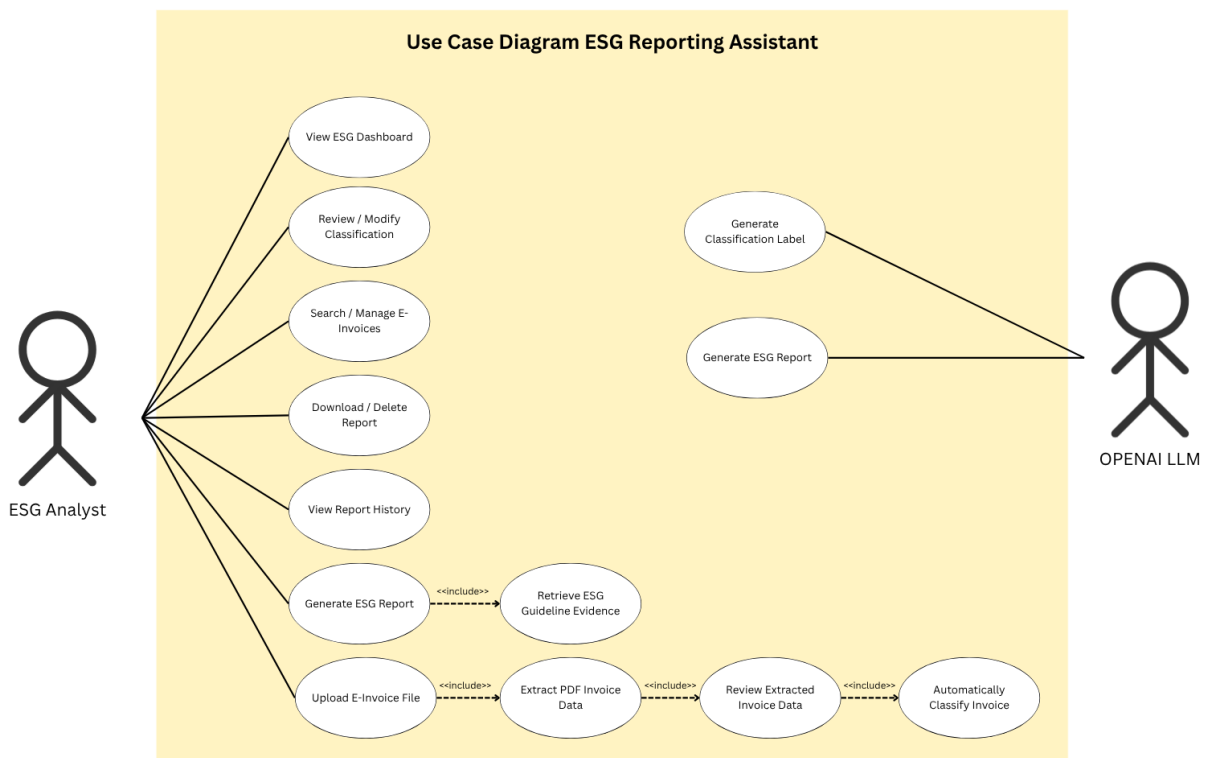


Figure 3.2.1 Use Case Diagram

Figure 3.2.1 presents the use case diagram for the ESG Reporting Assistant. The primary actor of the system was the ESG Analyst. The ESG Analyst interacted with the system to manage e-invoices, review ESG classifications, generate ESG reports, and access previously generated reports.

The ESG Analyst was able to view the ESG dashboard, which displayed a summary of invoice records, classification results, and report-related information. The analyst could also search and manage e-invoices stored in the system. This function supported the retrieval, review, and management of invoice data required for ESG analysis.

For invoice processing, the ESG Analyst uploaded an e-invoice file in PDF format. The system then extracted relevant invoice data from the uploaded document. The extracted information was presented to the analyst for review before the invoice was automatically classified. This review step enabled the analyst to verify the extracted data and reduce potential errors caused by incomplete or inaccurate PDF extraction.

The system generated classification labels by using the OpenAI LLM. The classification process identified ESG-related information from invoice content and assigned an appropriate category or label. The ESG Analyst could review and modify the classification result when necessary. This ensured that human judgement remained available for cases where the automatically generated classification was unsuitable or required further verification.

The ESG Analyst could generate an ESG report based on the available invoice records and classification results. During report generation, the system retrieved relevant ESG guideline evidence to support the generated content. The OpenAI LLM was used to produce structured ESG report content based on the classified data and retrieved guideline information. This approach improved the relevance of the report and supported traceability to ESG guidelines.

After a report had been generated, the ESG Analyst could view the report history to access previously generated reports. The analyst could also download a report for further use or delete a report that was no longer required. These functions allowed the system to provide complete report lifecycle management, from e-invoice upload and classification to report generation, review, and storage.

3.1.3 Activity Diagram

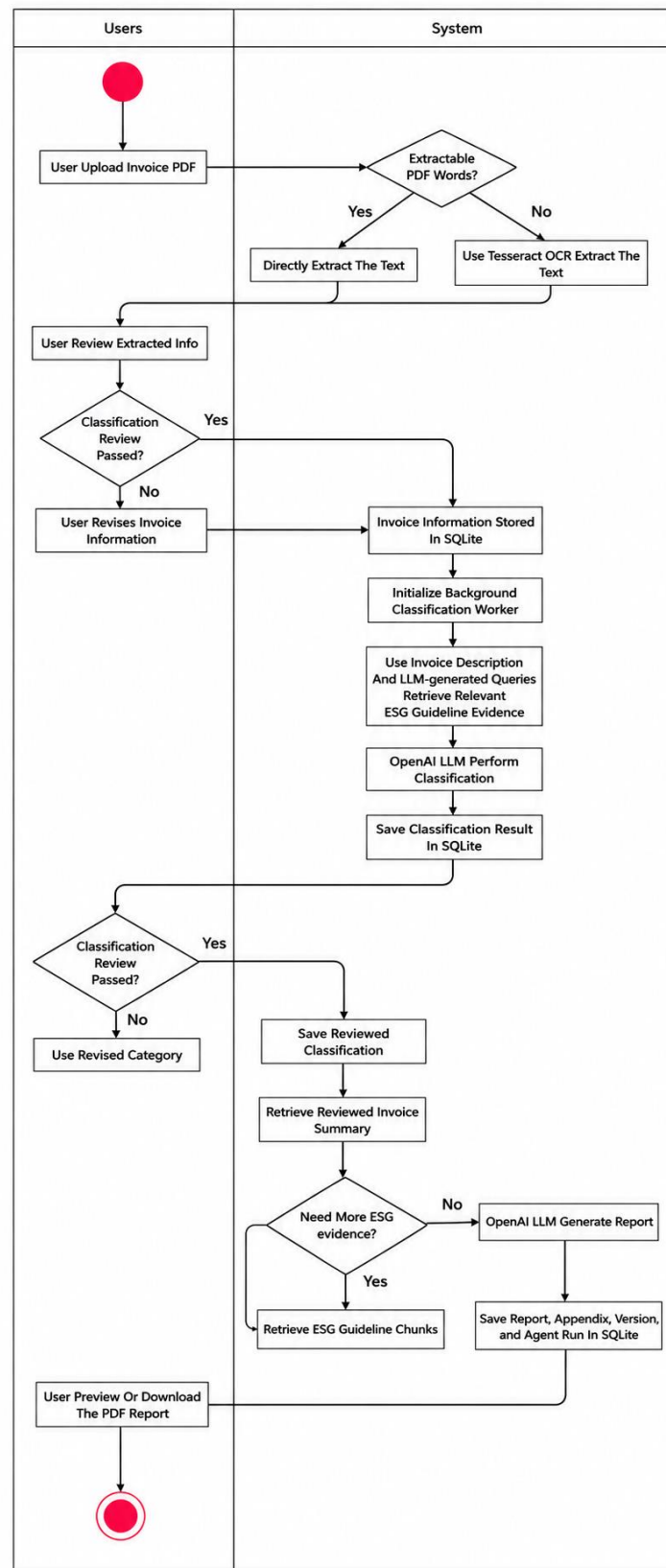


Figure 3.1.3 Activity Diagram

Figure 3.1.3 illustrates the workflow of the ESG Reporting Assistant. The process began when the user uploaded an e-invoice in PDF format. The system first determined whether the PDF contained extractable text. If extractable text was available, the system directly extracted the text from the document. Otherwise, Tesseract OCR was used to recognise and extract the text from the scanned invoice. The system then presented the extracted invoice information to the user for review and verification.

After reviewing the extracted information, the user determined whether the invoice information was correct. If the review was not passed, the user revised the invoice information before proceeding. Once the information was confirmed, the invoice information was stored in the SQLite database. The system then initiated a background classification worker to process the invoice without requiring the user to wait for the classification process to complete.

During the classification process, the system used the invoice description and LLM-generated queries to retrieve relevant ESG guideline evidence. The retrieved evidence was provided to the OpenAI LLM to perform ESG classification. The generated classification result was then saved in SQLite for subsequent review. The user reviewed the generated classification result and determined whether the classification was acceptable. If the classification was accepted, it was saved as the reviewed classification. Otherwise, the user revised the assigned ESG category before the reviewed classification was saved.

For ESG report generation, the system retrieved the summaries of the reviewed invoices. The system then determined whether additional ESG evidence was required. If more evidence was needed, relevant ESG guideline chunks were retrieved to provide additional supporting context. Once sufficient evidence was available, the reviewed invoice summaries and ESG guideline evidence were provided to the OpenAI LLM to generate the ESG report.

After the report had been generated, the system stored the report, appendix, version information, and agent run record in the SQLite database to support data persistence and traceability. Finally, the user could preview the generated ESG report or download it in PDF format. The workflow was then completed.

Chapter 4

SYSTEM DESIGN

4.1 System Block Diagram

4.1.1 Overall Block Diagram

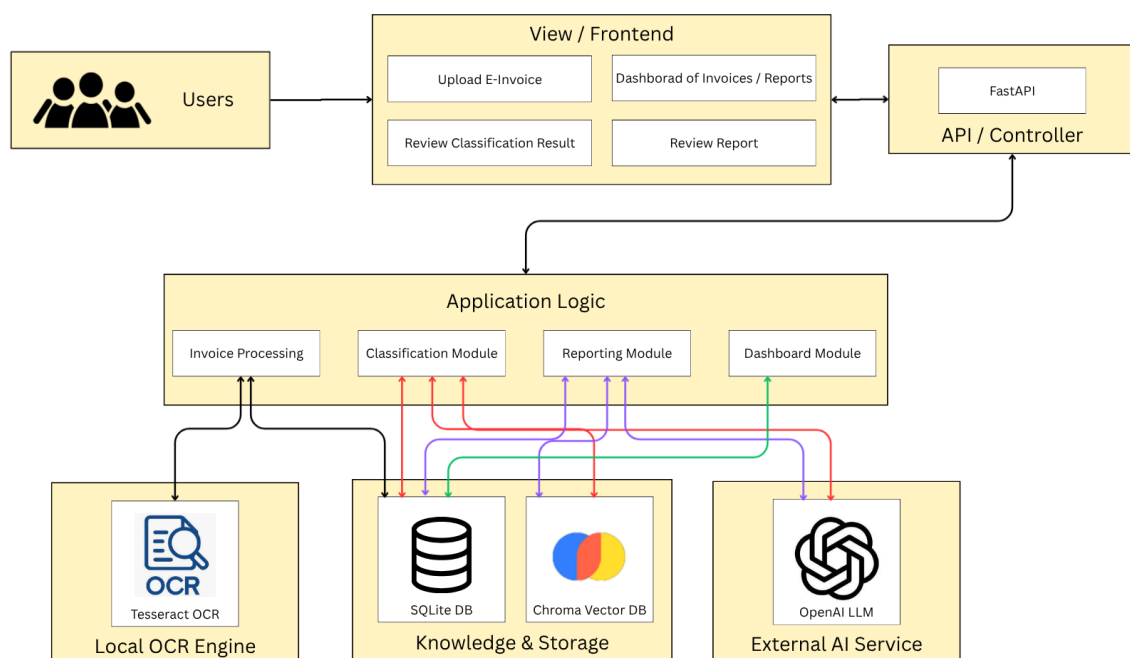


Figure 4.1.1 Overall Block Diagram

Figure 4.1.1 presents the overall block diagram of the ESG Reporting Assistant. The system was organised into five main parts: the View/Frontend, API/Controller, Application Logic, Knowledge and Storage, and external processing services. Users interacted with the system through the frontend to upload e-invoices, access the invoice and report dashboard, review classification results, and review generated reports. User requests were submitted to the FastAPI controller, which validated and routed them to the appropriate application module.

The application logic consisted of four main modules: Invoice Processing, Classification, Reporting, and Dashboard. The Invoice Processing Module handled uploaded e-invoices. It first attempted to extract embedded text from the PDF and used the local Tesseract OCR engine when sufficient text could not be extracted. The extracted invoice fields were presented to the user for verification before the verified invoice information was stored in the SQLite database.

The Classification Module analysed verified invoice descriptions through an evidence-grounded RAG workflow. Relevant ESG guideline chunks were retrieved from the Chroma vector database and supplied, together with the invoice information, to the OpenAI LLM. The model generated an Environmental, Social, Governance, or Non-ESG classification, accompanied by a brief reason and supporting evidence references. The classification result was stored in SQLite with a pending status and was subsequently presented to the user for human review.

The Reporting Module retrieved reviewed invoice classifications from SQLite and relevant ESG guideline evidence from Chroma. These inputs were provided to the OpenAI LLM to generate a structured invoice-based ESG draft report. The generated report and its supporting evidence information were stored for subsequent preview and download. The Dashboard Module retrieved invoice, classification, and report information from SQLite to produce summary cards, category totals, expenditure summaries, and recent processing records.

This architecture separated the user interface, request handling, business logic, local knowledge storage, document-processing services, and external AI service. SQLite served as the primary relational database for operational records, while Chroma provided semantic retrieval of ESG guideline content. Tesseract OCR was executed locally, whereas the OpenAI LLM was accessed as an external service for classification and report generation. This separation reduced component dependencies and supported the traceability of generated classifications and reports.

4.1.2 Invoice Processing and Classification Pipeline

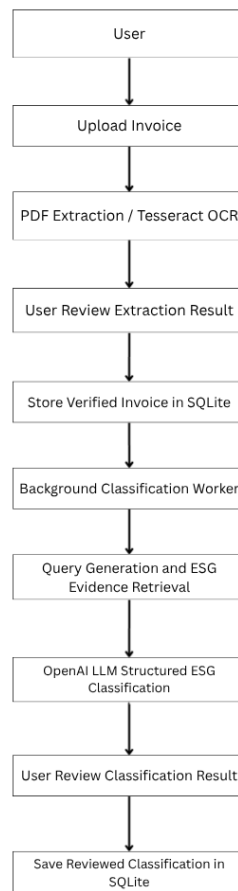


Figure 4.1.2 Invoice Processing and Classification Pipeline Block Diagram

Figure 4.1.2 illustrates the invoice processing and ESG classification workflow of the proposed system. The process begins when a user uploads an invoice document. The system first attempts to extract text directly from the PDF; when the PDF does not contain usable embedded text, Tesseract OCR is used as a fallback mechanism. The extracted invoice information is then presented to the user for review and correction. Only verified invoice data are stored in the SQLite database, which prevents inaccurate OCR or extraction results from being used in subsequent ESG analysis.

After the invoice is verified, a background classification worker processes the invoice asynchronously. The worker uses the invoice description to generate ESG-related search queries and retrieve relevant guideline evidence from the ESG knowledge base. The invoice details and retrieved evidence are then supplied to the OpenAI LLM, which produces a structured ESG classification result, including the ESG category, confidence level, justification, and supporting evidence references. The classification result is displayed to the

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user for human review. The user may confirm or revise the result before the reviewed classification is saved in SQLite. This workflow combines automated extraction and classification with human verification to improve the reliability, traceability, and auditability of ESG-related invoice records.

4.1.3 Agentic Report Generation Pipeline

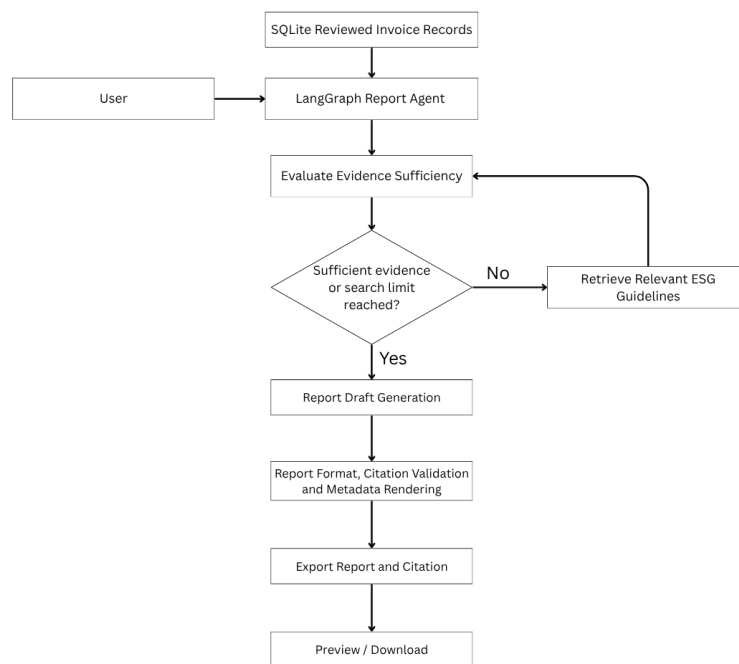


Figure 4.1.3 Agentic Report Generation Pipeline Block Diagram

Figure 4.1.3 illustrates the agentic report generation pipeline of the proposed system. The process starts when the user requests a report for a selected reporting period. The LangGraph Report Agent retrieves the reviewed invoice records from SQLite and uses them as the reporting snapshot. It then evaluates whether the available invoice information and ESG guideline evidence are sufficient to prepare an evidence-led report. If the evidence is insufficient and the search limit has not been reached, the agent retrieves additional relevant ESG guideline chunks from the local knowledge base. The retrieved evidence is then returned to the evidence-sufficiency evaluation stage, forming an iterative search loop. This loop

enables the agent to perform additional targeted searches when necessary while preventing unlimited retrieval attempts.

When sufficient evidence is available, or when the maximum search limit is reached, the system generates the report draft. The generated Markdown report is subsequently checked against the report format contract, including the required sections, invoice-based disclosure statement, performance table, and human-review actions. Invoice citations are validated against the reviewed invoice snapshot to ensure that cited invoices are valid, reviewed, and within the selected reporting period. For guideline citations, the system validates the internal guideline marker through an exact Chunk ID lookup in the vector database. It then renders user-readable citation metadata, including the source, topic, section, and page number, without exposing the technical Chunk ID in the public report. After validation, the system saves the draft report and its internal citation audit record before generating the public report export and Evidence Register. Finally, the user can preview or download the draft report for human review.

4.2 System Component Specifications

4.2.1 User Interaction and Application Service Layer

The User Interaction and Application Service Layer acts as the entry point between users and the underlying system services. It consists of the web-based presentation layer and the FastAPI REST API layer. The presentation layer allows users to upload e-invoices, review extracted invoice information, review ESG classification results, manage invoice and report records through the dashboard, and generate or download ESG reports.

The input to this component consists of user actions and requests, such as uploading a PDF invoice, confirming extracted information, revising a classification result, or requesting a report for a selected reporting period. The frontend sends these requests to the backend through REST API calls. The FastAPI application receives the requests, validates the required parameters, and forwards the request to the appropriate service component.

For example, an invoice upload request is forwarded to the invoice processing service, while a report generation request is forwarded to the reporting service and LangGraph report agent.

The API layer also retrieves processed results, invoice records, classifications, report history, and export files before returning them to the frontend for display. The frontend does not directly access the SQLite database, Chroma vector database, or external OpenAI service.

The output of this component is a user-friendly interface that displays processing status, extracted invoice details, ESG classifications, supporting evidence, and generated reports. It also provides users with review controls before important records are finalised.

The main technologies used in this layer are Next.js for the frontend interface and FastAPI for the backend REST API. This separation is important because it separates presentation logic from business logic and data access. It also ensures that sensitive operations, such as database access, LLM calls, evidence retrieval, and report validation, are controlled by the backend service layer rather than being exposed directly to the frontend.

4.2.2 Invoice Processing, OCR Fallback, and Human Review Component

The Invoice Processing, OCR Fallback, and Human Review Component is responsible for converting an uploaded invoice PDF into a verified invoice record that can be used for ESG classification and reporting. Its main responsibility is to extract relevant invoice information while ensuring that unverified or incorrectly extracted content is not used as evidence by subsequent components.

The input to this component is a PDF invoice uploaded by the user. The system first attempts to extract text directly from the PDF using native PDF text extraction. This method is preferred because it is faster and usually produces more accurate text when the PDF contains a selectable text layer.

However, some invoices may be scanned documents or image-based PDFs without extractable text. If native text extraction fails, returns insufficient text, or detects that the document is image-based, the system activates Tesseract OCR as a fallback method. Tesseract OCR converts the visible text in the invoice image into machine-readable text. The fallback design prevents the system from rejecting useful invoices merely because they are scanned rather than digitally generated.

After text extraction, the system prepares the extracted invoice information for user review. The user can review important fields such as invoice number, supplier, invoice date, description, amount, and extracted text. If the information is incomplete or incorrect, the user may correct it before confirmation. Only after user confirmation is the invoice stored as a verified invoice record in SQLite.

Once a verified record is created, the system may create a background classification job. This allows invoice classification to be performed without forcing the user to wait on the upload page. The classification result is later returned to the user for another review stage.

The output of this component is a verified invoice record containing the reviewed invoice details and extracted textual content. This verified record becomes the input for the classification component. The use of human review is an important design decision because OCR and PDF extraction may produce errors, especially for scanned documents, unclear images, complex invoice layouts, or low-resolution files. The review stage ensures that potentially incorrect extraction results are not automatically treated as factual business evidence.

The main technologies used are PDF text extraction libraries, Tesseract OCR, FastAPI background processing, and SQLite. The component communicates with the application service layer to receive uploaded files, with SQLite to store verified invoices, and with the classification component to initiate the next stage of processing.

4.2.3 ESG Knowledge Base and Evidence Retrieval Component

The ESG Knowledge Base and Evidence Retrieval Component provides the evidence foundation for ESG classification and report generation. Its responsibility is to store ESG guideline content in a searchable form and retrieve relevant guideline evidence when the system needs to classify an invoice or support a report statement.

The input to this component consists of ESG guideline documents, such as sustainability or disclosure standards. Before the documents can be retrieved efficiently, they are divided into smaller text chunks. Each chunk is stored together with metadata that describes its source and

location. The metadata includes the internal chunk identifier, document source, topic, section, page number or page range, source version, and index version.

The text chunks and their embeddings are stored in the Chroma vector database. When an invoice is being classified, the system uses the reviewed invoice description and other invoice information to construct a retrieval query. For more complex or vague invoice descriptions, the LLM may generate or rewrite the query so that it better reflects ESG-related concepts and can retrieve more relevant guideline content.

For example, an invoice that contains the description “electricity charges” may lead the system to retrieve guideline chunks related to energy consumption, energy management, or environmental disclosure. The retrieved chunks provide contextual evidence for the classification component. Similarly, the reporting agent can retrieve guideline chunks when it needs additional evidence to support a report statement.

The output of this component is a collection of relevant ESG guideline chunks with their associated metadata. The retrieved evidence is not treated as a final answer by itself. Instead, it is passed to the classification component or report agent as grounded reference material.

A key design feature is the use of metadata. Each guideline chunk has an internal `chunk_id`, which allows the system to identify the exact chunk used during retrieval and validation. However, the chunk ID is a technical identifier and is not intended for end users. It is used internally for exact lookup, traceability, and validation. Public reports display meaningful citation metadata, such as the guideline source, topic, section, and page number, instead of exposing the internal chunk ID.

The main technologies used are Chroma vector database, embedding-based similarity retrieval, metadata filtering, and the OpenAI LLM for query generation or query rewriting. This component communicates with the classification component and the LangGraph report agent. The use of a vector knowledge base is justified because ESG guidelines can be lengthy and contain many related topics. Retrieval allows the system to use only the most relevant evidence instead of passing the entire guideline document to the LLM.

4.2.4 Evidence-Grounded Invoice Classification Component

The Evidence-Grounded Invoice Classification Component determines whether a verified invoice is related to Environmental, Social, Governance, or Non-ESG activities. Its responsibility is not only to assign a category, but also to provide evidence that explains why the category was selected.

The input to this component consists of a verified invoice record from SQLite and the relevant ESG guideline evidence retrieved from the knowledge base. The invoice record may include the supplier name, invoice description, invoice amount, invoice date, and reviewed extracted text. The retrieved evidence provides ESG guideline context that helps the LLM make a more defensible classification decision.

The classification process begins when the background classification worker receives a verified invoice record. The system generates a retrieval query based on the invoice description and relevant invoice fields. If necessary, the query is rewritten to improve evidence retrieval. The system then sends the verified invoice information and retrieved guideline evidence to the OpenAI LLM.

The LLM is required to return a structured ESG classification result using a strict JSON schema. The structured response may include the ESG category, subcategory, explanation, confidence level, and evidence references. Using a strict schema prevents the LLM from returning an unstructured narrative that is difficult for the system to validate or store.

After receiving the LLM response, the system validates the JSON structure and verifies that the returned category belongs to the permitted classification values. It also checks that referenced evidence is available and that the response contains the required fields. If the output is invalid, incomplete, or unsupported, it can be rejected or marked for review rather than being saved as a trusted classification.

The classification result is then presented to the user. The user can review the suggested category, reason, and evidence before accepting the result or revising the category. Once confirmed, the reviewed classification is stored in SQLite and becomes available for dashboard summaries and report generation.

The output of this component is a reviewed ESG classification record with supporting evidence. This design is evidence-grounded because the LLM does not classify an invoice only from its own general knowledge. Instead, it receives relevant ESG guideline evidence retrieved from the system knowledge base. The additional human review step provides control over decisions that may affect reporting outcomes.

The main technologies used are FastAPI background workers, OpenAI LLM structured output, JSON schema validation, Chroma vector retrieval, and SQLite. This component communicates with the invoice processing component to receive verified invoices, the knowledge base component to retrieve evidence, and the reporting component to provide reviewed classification records.

4.2.5 Agentic Reporting, Citation Validation, and Export Component

The Agentic Reporting, Citation Validation, and Export Component is responsible for generating an ESG report based on reviewed invoice records and classifications. It uses a LangGraph report agent to coordinate evidence evaluation, guideline retrieval, report drafting, citation validation, and export generation.

The input to this component consists of reviewed invoice records and reviewed ESG classification results stored in SQLite. The user selects a reporting period and requests report generation through the frontend. The report agent retrieves only the reviewed invoice records that belong to the selected period. This ensures that draft reports are based on human-reviewed records rather than unverified extraction results or unreviewed classification outputs.

The LangGraph report agent first evaluates whether the available invoice records and evidence are sufficient to support the intended report content. If the current evidence is insufficient, the agent retrieves relevant ESG guideline chunks from the knowledge base. It then evaluates the evidence again. This process forms a controlled search loop.

The loop continues until the agent determines that the evidence is sufficient or reaches a predefined search limit. The search limit prevents the agent from repeatedly retrieving evidence without producing a report. The evidence sufficiency decision does not mean that the system invents missing ESG performance data. If the available invoices only indicate spending or

procurement activity, the report must present them as invoice-based observations and should not claim verified ESG performance outcomes that are not supported by the evidence.

After the evidence collection stage, the LLM generates a Markdown draft report. The report may include invoice-supported statements, report-period summaries, limitations, and guideline-supported disclosure context. Invoice citations are represented using invoice numbers or invoice-group references, such as [INV:INV-001] or [INV-GRP:E-01].

For guideline citations, the LLM uses internal markers such as {{GUIDELINE:chunk-id}} during the drafting process. These markers are not intended to appear in the final public report. The Report Contract and Citation Validation process checks whether each internal guideline citation marker refers to an allowed and retrievable guideline chunk.

The system performs an exact lookup using the chunk ID in the Chroma vector database. It then verifies the associated metadata, including the source, topic, section, page number, source version, and index version. If the chunk cannot be found, is not part of the retrieved evidence, or does not satisfy the required report contract, the citation is rejected or the draft is marked as invalid.

After validation, the internal marker is replaced with reader-friendly citation metadata. For example, an internal marker may be rendered as:

(Source: GRI 302; Topic: Energy; Section: Energy Consumption; Page: 5)

Therefore, the public report can provide traceable guideline references without displaying internal technical identifiers. The raw chunk ID remains stored only in the internal audit record together with the validation status, source metadata, report version, and supporting context.

The final output of this component includes a public PDF report, an Evidence Register CSV file, and an export package such as a ZIP file. The public PDF presents readable invoice and guideline citations. The Evidence Register provides a structured record of invoice references, classification evidence, validation status, and report version information. Users can preview the report before downloading the final export files.

The main technologies used are LangGraph for report-agent orchestration, OpenAI LLM for draft generation, Chroma vector database for exact guideline chunk validation, SQLite for

reviewed records and report audit history, Markdown rendering, PDF generation, and CSV export. This component communicates with SQLite to retrieve reviewed invoice records, with Chroma to retrieve and validate guideline evidence, and with the application service layer to return report results to the user.

The design of this component ensures that report generation is not a one-step LLM output process. Instead, the report is generated through a controlled workflow that includes reviewed data selection, evidence sufficiency evaluation, bounded guideline retrieval, citation validation, metadata rendering, and export generation.

4.3 Modules and Sub-modules of the Proposed System

4.3.1 Invoice Management Module

Sub-module	Description	Output
Invoice Upload	Allows users to upload invoice documents in PDF format for processing.	Uploaded invoice file
Invoice Text Extraction and OCR Fallback	Extracts embedded text from the PDF. If native extraction is unavailable or insufficient, Tesseract OCR is used to process scanned or image-based invoices.	Extracted invoice text
Extraction Review	Allows users to verify and correct extracted invoice details, including invoice number, supplier, date, description, and amount.	Verified invoice information
Invoice Record Management	Stores the verified invoice record in SQLite and prepares it for classification processing.	Verified invoice record

Table 4.3.1: Sub-modules of the Invoice Management Module

The Invoice Management Module allows users to submit, verify, and manage invoice records. This module is the starting point of the system workflow because an invoice must be reviewed before it can be used for ESG classification or report generation.

The first sub-module is the **Invoice Upload Sub-module**. It allows the user to upload an invoice in PDF format. Once the invoice has been uploaded, the system creates an invoice processing request and begins extracting the textual content from the document.

The second sub-module is the **Invoice Text Extraction Sub-module**. The system first attempts to extract embedded text directly from the PDF. If the uploaded document is scanned or does not contain sufficient extractable text, the system uses Tesseract OCR as a fallback method. This enables the system to process both digitally generated invoices and image-based scanned invoices.

The third sub-module is the **Extraction Review Sub-module**. The user is shown the extracted invoice details, such as the invoice number, supplier, date, description, and amount. The user can verify or correct the extracted information before saving the invoice. This review stage ensures that inaccurate extraction or OCR results are not directly used by subsequent AI functions.

The final sub-module is the **Invoice Record Management Sub-module**. After the user confirms the extracted information, the invoice is stored as a verified invoice record in SQLite. The user can later view the invoice record and its processing status through the dashboard. A verified invoice may then be sent to the background classification process.

4.3.2 ESG Classification and Review Module

Sub-module	Description	Output
Background Classification	Creates and processes a background classification job after an invoice has been verified.	Classification job status
Query Generation and ESG Evidence Retrieval	Generates or rewrites a query from invoice information and retrieves relevant ESG guideline evidence from the knowledge base.	Relevant ESG guideline evidence
Structured ESG Classification	Uses the OpenAI LLM to generate a structured classification result containing the ESG category, explanation, confidence, and evidence references.	Proposed ESG classification
Classification Review	Allows users to accept or revise the suggested ESG classification before it is finalised.	Reviewed classification result

Sub-module	Description	Output
Classification Record Storage	Stores the reviewed classification and its evidence information in SQLite for dashboard and reporting use.	Saved reviewed classification record

Table 4.3.2: Sub-modules of the ESG Classification and Review Module

The ESG Classification and Review Module determines the ESG relevance of a verified invoice and allows the user to review the generated classification result. The module is designed to support evidence-grounded classification instead of assigning a category based only on the invoice description.

The first sub-module is the **Background Classification Sub-module**. Once an invoice has been verified, the system creates a classification job. The classification process is performed in the background so that users do not need to remain on the upload page while the system retrieves evidence and generates a result.

The second sub-module is the **ESG Evidence Retrieval Sub-module**. The system uses the verified invoice description and related invoice information to generate a search query. It then retrieves relevant ESG guideline chunks from the Chroma vector database. If required, the query may be rewritten to improve the relevance of the retrieved guideline evidence.

The third sub-module is the **Structured ESG Classification Sub-module**. The system sends the verified invoice information and retrieved ESG evidence to the OpenAI LLM. The LLM returns a structured result containing the suggested ESG category, explanation, confidence level, and evidence references. The supported categories are Environmental, Social, Governance, and Non-ESG.

The fourth sub-module is the **Classification Review Sub-module**. The user reviews the suggested classification result and its supporting evidence. The user may accept the recommended category or revise it when necessary. Once confirmed, the reviewed classification is saved in SQLite and becomes available for reporting and dashboard display.

This module ensures that the final classification is based on verified invoice data, relevant ESG guideline evidence, system validation, and human review.

4.3.3 ESG Report Generation Module

Sub-module	Description	Output
Report Request and Record Selection	Retrieves reviewed invoice records and classifications within the reporting period selected by the user.	Report input records
Evidence Sufficiency Evaluation	The LangGraph report agent evaluates whether the available evidence is sufficient for reporting. If necessary, it retrieves additional relevant ESG guidelines until sufficient evidence is obtained or the search limit is reached.	Validated evidence set or search-limit result
Report Draft Generation	Generates a Markdown ESG report draft using reviewed invoice records, reviewed classifications, and retrieved guideline evidence.	Draft report
Citation Validation and Metadata Rendering	Validates invoice references and guideline citation markers. Internal guideline chunk IDs are resolved into public source, topic, section, and page metadata.	Validated report content with public citations
Report Export	Generates the public PDF report, Evidence Register CSV file, and downloadable report package.	Report export files

Table 4.3.3: Sub-modules of the ESG Report Generation Module

The ESG Report Generation Module enables users to generate an ESG report based on reviewed invoice records and reviewed classification results. This module uses the LangGraph report agent to coordinate evidence retrieval, report drafting, validation, and export.

The first sub-module is the **Report Request and Record Selection Sub-module**. The user selects a reporting period and requests a report through the system interface. The system retrieves only reviewed invoice records and classifications that fall within the selected reporting period. This prevents unverified invoices or unreviewed classifications from being included in the report.

The second sub-module is the **Evidence Sufficiency Evaluation Sub-module**. The LangGraph report agent evaluates whether the available invoice information and ESG evidence are sufficient to support the report content. If the evidence is insufficient and the search limit has not been reached, the agent retrieves additional relevant ESG guideline chunks from the knowledge base. The agent then evaluates the evidence again before proceeding.

The third sub-module is the **Report Draft Generation Sub-module**. Once sufficient evidence has been collected, or the permitted search limit has been reached, the system generates a Markdown draft report using the reviewed invoice records, classifications, and retrieved ESG guideline evidence. The report may include invoice-based observations, ESG category summaries, limitations, and relevant disclosure context.

The fourth sub-module is the **Citation Validation and Metadata Rendering Sub-module**. The report draft may contain invoice references and internal guideline citation markers. The system validates invoice references against the reviewed invoice records. For guideline citations, the system performs an exact lookup of the internal chunk ID and verifies its source, topic, section, page number, and version metadata. The internal marker is then replaced with a reader-friendly citation, such as the guideline source, topic, section, and page number.

The final sub-module is the **Report Export Sub-module**. After validation, the system generates a public PDF report, an Evidence Register CSV file, and a downloadable report package. The user can preview the report before downloading the final files.

4.3.4 Dashboard and Record Management Module

Sub-module	Description	Output
Invoice and Classification Dashboard	Displays invoice extraction, verification, review, and classification statuses.	Dashboard summary
Filtering and Record Retrieval	Allows users to search and filter stored invoice and classification records using criteria such as supplier, date, category, status, or reporting period.	Filtered records

Sub-module	Description	Output
Report History	Displays previously generated reports and allows users to retrieve related exports.	Report history and downloads

Table 4.3.4: Sub-modules of the Dashboard and Record Management Module

The Dashboard and Record Management Module provides users with an overview of invoices, ESG classifications, and generated reports. It allows users to review system records without needing to repeat the entire processing workflow.

The first sub-module is the **Invoice and Classification Dashboard Sub-module**. It displays invoice records together with their extraction, verification, and classification status. Users can identify whether an invoice is pending review, classified, reviewed, or ready for reporting.

The second sub-module is the **Filtering and Record Retrieval Sub-module**. Users can search and filter invoice records based on selected criteria, such as invoice date, supplier, ESG category, review status, or reporting period. This helps users locate relevant invoice records when reviewing classifications or preparing a report.

The third sub-module is the **Report History Sub-module**. It displays previously generated reports and their related metadata, such as the reporting period, report version, generation date, and validation status. Users can retrieve earlier reports and download their associated export files when needed.

The dashboard module improves usability by presenting the results of the invoice processing, classification, and reporting modules in a single location. It also supports traceability because users can review how an invoice progresses from upload, verification, classification, and finally inclusion in a report.

4.4 Concluding Remark

This chapter presented the design of the proposed ESG invoice classification and reporting system. The system architecture, component specifications, and functional modules were described to illustrate how invoice data is extracted, reviewed, classified using ESG guideline evidence, and transformed into validated ESG reports. The next chapter discusses the implementation of the proposed system.

Chapter 5

SYSTEM IMPLEMENTATION

5.1 Hardware Setup

Component	Specification
Laptop model	HP Laptop 15s-eq3xxx
Processor	AMD Ryzen 5 5625U with Radeon Graphics, 2.30 GHz, 6 cores, 12 logical processors
Memory	8 GB RAM
Operating system	Microsoft Windows 11 Home Single Language, 64-bit
Graphics	AMD Radeon Graphics (integrated)

Table 5.1.1: Hardware specifications used for system development and testing.

The ESG Invoice Classification and Reporting System was developed and tested on an HP Laptop 15s-eq3xxx. The computer was equipped with an AMD Ryzen 5 5625U processor with integrated Radeon Graphics, 8 GB of RAM and Microsoft Windows 11 Home Single Language 64-bit operating system.

The hardware environment supported the local execution of the Next.js frontend, FastAPI backend, SQLite database, Chroma vector database and CPU-based embedding model during development.

5.2 Software Setup

Component	Specification
Frontend	Next.js and TypeScript
Backend	FastAPI and Python
Database	SQLite, Chroma DB
Embedding model	sentence-transformers / MiniLM
OCR	Tesseract OCR
LLM integration	LangChain and OpenAI API
Workflow Orchestration	LangGraph

Table 5.1.1: Hardware specifications used for system development and testing.

The ESG Invoice Classification and Reporting System was implemented as a web-based application. The frontend was developed using Next.js and TypeScript to provide interfaces for invoice import, classification review, dashboard visualisation, report generation and report download. The backend was developed using FastAPI and Python to provide REST API endpoints and to coordinate invoice processing, classification, ESG evidence retrieval and report generation. SQLite was used as the local operational database to store invoices, classification records, background job states, generated reports and evidence-register snapshots.

For the artificial intelligence and retrieval functions, sentence-transformers with a MiniLM embedding model was used to generate local vector representations of ESG guideline documents and search queries. Chroma was used to store and retrieve the embedded guideline chunks. Tesseract OCR was used as a fallback mechanism when invoice text could not be extracted directly from PDF files. LangChain integrated the language model and retrieval components, while LangGraph orchestrated the stateful background report-generation workflow. The OpenAI API, using the GPT-4o mini model, was used to perform structured ESG invoice classification and evidence-led report generation.

5.3 System Configuration and Backend Integration

5.3.1 Local Data Persistence and Invoice Processing

The system used SQLite as its local persistent database. SQLite was selected because the project was implemented as a self-contained web-based prototype and did not require a separate database server. The database retained the operational data required by the system, including invoice records, invoice versions, ESG classification results, human-review status, background job information and generated report records.

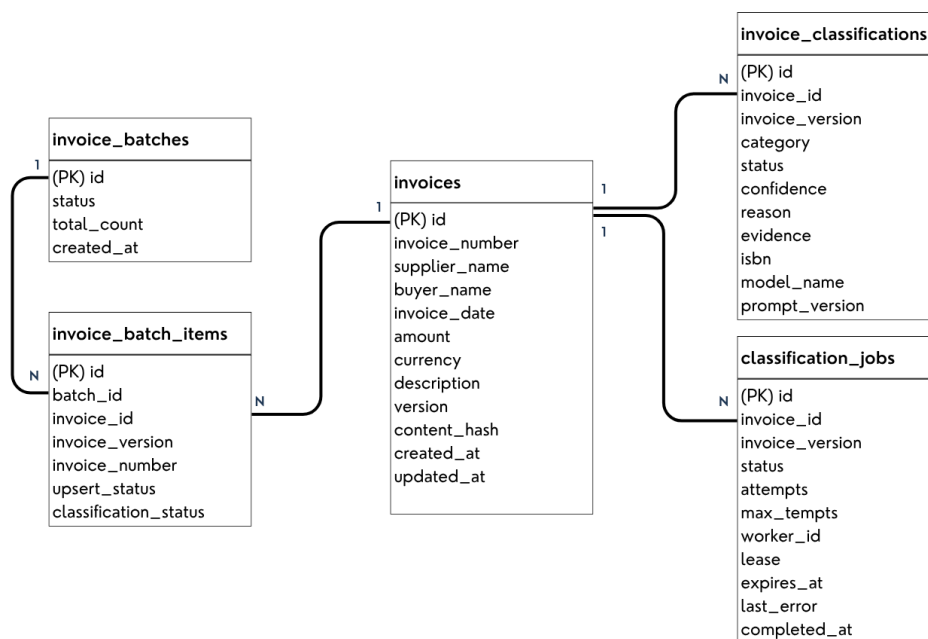


Figure 5.3.1: Entity relationship diagram for invoice persistence and classification processing.

Figure 5.3.1 illustrates the simplified entity relationship diagram for invoice persistence and classification processing. The `invoice_batches` table stores the overall status and summary of each invoice-import batch. Each batch contains multiple records in `invoice_batch_items`, which record the processing position, upsert result and classification status of individual imported invoices.

The `invoices` table acts as the central entity and stores extracted invoice information, including the invoice number, supplier, date, amount, currency, description and version number. An invoice may have multiple classification records in `invoice_classifications`, allowing the ESG category, confidence level, explanation and retrieved guideline evidence to be retained for

different invoice versions. The `classification_jobs` table stores the asynchronous worker job associated with an invoice version, including its processing status, retry attempts, worker identifier and error information. This design supports traceable invoice processing, background classification and subsequent human review.

5.3.2 ESG Guideline Retrieval Configuration

The ESG guideline retrieval module was configured to provide local and traceable guideline evidence for invoice classification and ESG report generation. The Bursa Malaysia Sustainability Reporting Guide was retained locally within the project and indexed into the system's vector database. This approach enabled the system to retrieve relevant ESG disclosure requirements without depending on an external web service during normal operation.

Figure 5.3.2 shows an example of the original guideline content. The document was organised into Environmental, Social and Governance sections, followed by individual disclosure topics and bullet-point requirements. For example, the Environmental section included topics such as core environmental disclosures, energy and emissions, water and waste, and environmental management approach. This structured format was used to preserve the context of each guideline requirement during indexing.

Bursa Malaysia Sustainability Reporting Guide

Environmental (E)

Core environmental disclosures

- Environmental reporting should focus on material impacts, management approach, indicators, and performance data.
- Where climate is material, Main Market issuers should provide climate-related disclosures aligned with TCFD recommendations.
- ACE Market corporations should disclose a basic transition plan toward a low-carbon economy.
- Environmental indicators should be reported consistently across periods using stable definitions, assumptions, and calculation methods.
- At least three financial years of data should be disclosed on a rolling basis for reported environmental indicators.

Energy and emissions

- Energy management indicators should include total energy consumption.
- Energy data should use clear units and a consistent calculation basis across reporting periods.
- Emissions indicators should include Scope 1 and Scope 2 emissions.
- The guide also expects at least selected Scope 3 emissions, such as business travel and employee commuting.
- If climate targets are set, the company should disclose the base year, target year, and progress against target.

Water and waste

- Water indicators should include total volume of water used.
- Waste management indicators should include total waste generated.
- Waste disclosures should distinguish waste diverted from disposal and waste directed to disposal.
- Scoping choices should not exclude facilities or activities with significant environmental impact.
- If environmental data is restated, the company should explain the reason for the restatement.

Figure 5.3.2: Example of structured content in the Bursa Malaysia Sustainability Reporting Guide.

During indexing, the system extracted text from the guideline PDF page by page and identified the ESG section, topic and related bullet points. Related bullet points under the same ESG section and topic were combined into a guideline chunk. Each chunk contained a maximum of three related bullet points and was created only when the combined content contained at least ten words. Each chunk was assigned a unique identifier and retained metadata consisting of its source file name, ESG section, topic and page reference. Each guideline chunk was converted into a vector representation using a locally bundled Sentence-Transformers embedding model based on *MiniLM-L6-H384-uncased*. The embedding model was stored locally with the backend and loaded in local-only mode. Therefore, the system did not need to download the model from Hugging Face or depend on an internet connection during indexing and retrieval.

The same embedding model was used to convert a user query into a vector representation,

enabling both the query and guideline chunks to be compared within the same semantic vector space. The resulting vectors and their associated metadata were persisted in the local Chroma vector database.

When a user's invoice information or a report-generation task required ESG evidence, the system converted the query into an embedding and performed semantic similarity retrieval against the Chroma collection. Figure 5.3.3 presents an example of a retrieved evidence chunk for the query related to health and safety indicators. The result was retrieved from *Bursa_ESG_reordered_E_S_G.pdf*, under the Social (S) section and the *Health, safety, and community* topic on page 2. The retrieved text specified relevant health and safety indicators, including work-related fatalities, lost time incident rate and employee training on health and safety standards.

```
-----  
Chunk ID : Bursa_ESG_reordered_E_S_G.pdf_c12  
Source   : Bursa_ESG_reordered_E_S_G.pdf  
Section  : Social (S)  
Topic    : Health, safety, and community  
Page     : 2  
Evidence :  
Health and safety indicators should include work-related fatalities. Health and safety  
indicators should include lost time incident rate. Health and safety indicators  
should include employees trained on health and safety standards.  
=====
```

Figure 5.3.3: Example of a retrieved ESG guideline evidence chunk from the local Chroma vector database.

The retrieved result preserved its chunk ID and source metadata together with the supporting text. This ensured that ESG classification and report statements could be linked back to the relevant guideline topic and original page. Consequently, the RAG pipeline provided not only semantically relevant information, but also evidence traceability for user review and report generation.

5.3.3 Background Worker and Report-Agent Integration

The system integrated two long-running server-side processes: the background invoice-classification worker and the agentic report-generation workflow. Both processes were designed to execute independently from the user interface. As a result, users could continue

navigating to other pages while invoice classification or report generation remained active in the backend.

When an invoice was imported or its version was updated, the system created a record in the `classification_jobs` table. Each job was associated with a specific invoice ID and invoice version, ensuring that classification results could be traced to the correct version of the invoice. When the backend started, it checked for unclassified invoices, recovered any interrupted jobs and started the configured classification workers. Each worker atomically claimed one queued job, changed its status to running and then performed the classification process using the invoice information and retrieved ESG guideline evidence.

The classification-job design included a lease and retry mechanism to improve reliability. A worker refreshed its lease while processing a job, allowing interrupted work to be identified if the lease expired. If an error occurred, the job was returned to the queue with a delayed retry. The system allowed a maximum of three attempts; a job that continued to fail was marked as failed and its error information was retained. On successful completion, the classification result and the related import-batch item status were updated in SQLite. This persistent job state allowed the frontend to display classification progress without requiring the user to remain on the invoice-import page.

Report generation was implemented as a separate background LangGraph workflow. When the user selected a reporting period and started report generation, the system created a report-agent run with an initial queued status and executed the workflow in a dedicated background thread. Before the agent began its reasoning process, the system loaded only invoices with a Reviewed classification status within the selected reporting period. These invoices were stored as a fixed snapshot in the agent state. Therefore, any later invoice changes did not affect the evidence used by an already running report-generation task.

The LangGraph workflow followed the sequence: *prepare reviewed invoice snapshot* → *agent decision* → *one local guideline retrieval* → *agent decision*. At every decision step, the LLM either generated the report using the available evidence or requested one focused local ESG guideline search. The system permitted only one retrieval request per decision step, so the agent could assess the result of each search before deciding whether another search was necessary. To control execution cost and duration, the report agent was limited to a maximum of eight local ESG guideline searches.

After the agent produced report content, the system validated the report contract, invoice-group citations and guideline citations. Invoice references were checked against the reviewed invoice snapshot, while guideline citations were checked against the exact chunk IDs that had been retrieved or were already associated with the selected invoices. The validated report, evidence-register snapshot and citation records were then saved to the local database. Agent-run status and individual workflow steps were also persisted, while LangGraph checkpoints were stored in a separate local SQLite checkpoint file. The Generate Report page retrieved the current run state repeatedly and stored the active run ID in browser local storage. This enabled the agent-state window to show progressive retrieval results and to restore the running report task when the user returned to the page.

5.4 System Operation

5.4.1 Invoice Classification Workflow

The invoice-classification workflow began at the Dashboard page. As shown in Figure 5.4.1, the dashboard provided an overview of invoices within the selected reporting year. It displayed the total number of invoices and the corresponding expenditure amount, together with the number and amount classified under the Environmental, Social and Governance categories. The page also displayed recent invoice classifications, allowing users to review the latest classification status and category assignments.

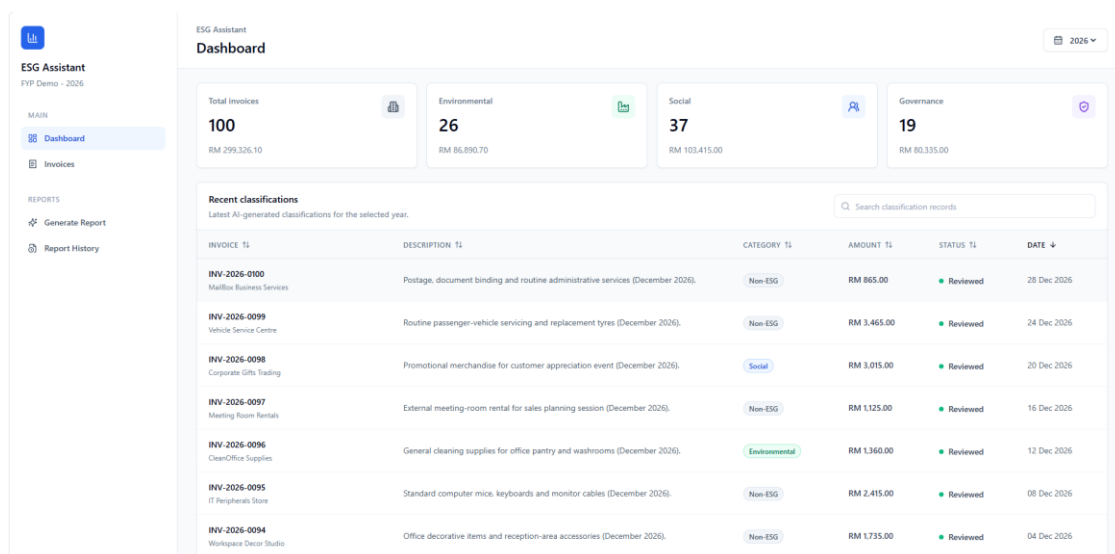


Figure 5.4.1: Dashboard overview of invoice classifications and ESG category summaries.

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To manage invoice records, users navigated to the Invoices page. Figure 5.4.2 shows the invoice list, which displayed the invoice number, supplier, date, description, predicted category, amount and classification status. The page also provided filters for all invoices, pending invoices and reviewed invoices. Users selected the **Upload invoice** function to begin importing a new invoice.

The screenshot shows the 'Invoices' page in the ESG Assistant system. The left sidebar contains navigation links for 'MAIN' (Dashboard, Invoices) and 'REPORTS' (Generate Report, Report History). The main content area displays a table of invoice records. Above the table, there are filters for 'All 100', 'Pending 0', and 'Reviewed 100', along with buttons for 'Upload invoice' and 'Advanced filter'. The table has columns for 'INVOICE T1', 'DATE', 'DESCRIPTION', 'CATEGORY', 'AMOUNT', and 'STATUS'. All 10 records shown are marked as 'Reviewed'.

INVOICE T1	DATE	DESCRIPTION	CATEGORY	AMOUNT	STATUS
INV-2026-0100	28 Dec 2026	Postage, document binding and routine administrative services (December 2026).	Non-ESG	RM 865.00	Reviewed
INV-2026-0099	24 Dec 2026	Routine passenger-vehicle servicing and replacement tyres (December 2026).	Non-ESG	RM 3,465.00	Reviewed
INV-2026-0098	20 Dec 2026	Promotional merchandise for customer appreciation event (December 2026).	Social	RM 3,015.00	Reviewed
INV-2026-0097	16 Dec 2026	External meeting-room rental for sales planning session (December 2026).	Non-ESG	RM 1,125.00	Reviewed
INV-2026-0096	12 Dec 2026	General cleaning supplies for office pantry and washrooms (December 2026).	Environmental	RM 1,360.00	Reviewed
INV-2026-0095	08 Dec 2026	Standard computer mice, keyboards and monitor cables (December 2026).	Non-ESG	RM 2,415.00	Reviewed
INV-2026-0094	04 Dec 2026	Office decorative items and reception-area accessories (December 2026).	Non-ESG	RM 1,735.00	Reviewed
INV-2026-0093	30 Nov 2026	Domestic courier charges for routine customer deliveries (November 2026).	Social	RM 1,065.00	Reviewed
INV-2026-0092	26 Nov 2026	Routine business-card and brochure printing service (November 2026).	Non-ESG	RM 1,455.00	Reviewed

Figure 5.4.2: Invoice management page with invoice-upload function.

After an invoice PDF was uploaded, the system displayed an extraction-confirmation window before saving the record. As shown in Figure 5.4.3, the window presented the extracted invoice number, supplier, buyer, invoice date, amount, currency and description, together with the complete extracted text from the source file. For a PDF containing selectable text, the system performed direct PDF-text extraction. If the PDF did not contain sufficient extractable text, Tesseract OCR was used as a fallback mechanism. The extracted invoice fields were editable; therefore, users could correct inaccurate values before selecting **Confirm import**. This verification step reduced the risk of incorrect information being passed to the subsequent AI-classification process.

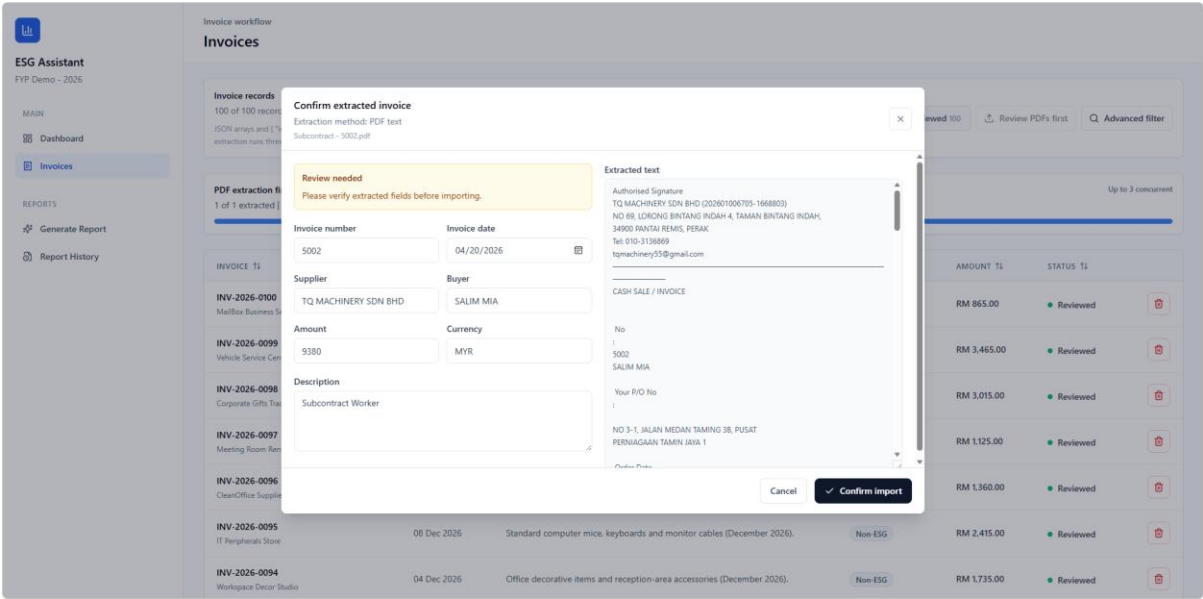


Figure 5.4.3: Invoice extraction preview and confirmation window.

Once the invoice was confirmed and saved, the system created a background classification job for the relevant invoice version. While the job was being processed, the invoice detail view displayed **Awaiting AI**. As illustrated in Figure 5.4.4, the imported invoice was shown with a Pending status in the invoice list. In this system, Pending indicated that the invoice had not yet received final human confirmation. The background worker processed the invoice independently from the user interface, allowing users to continue viewing other pages while the classification task was running.

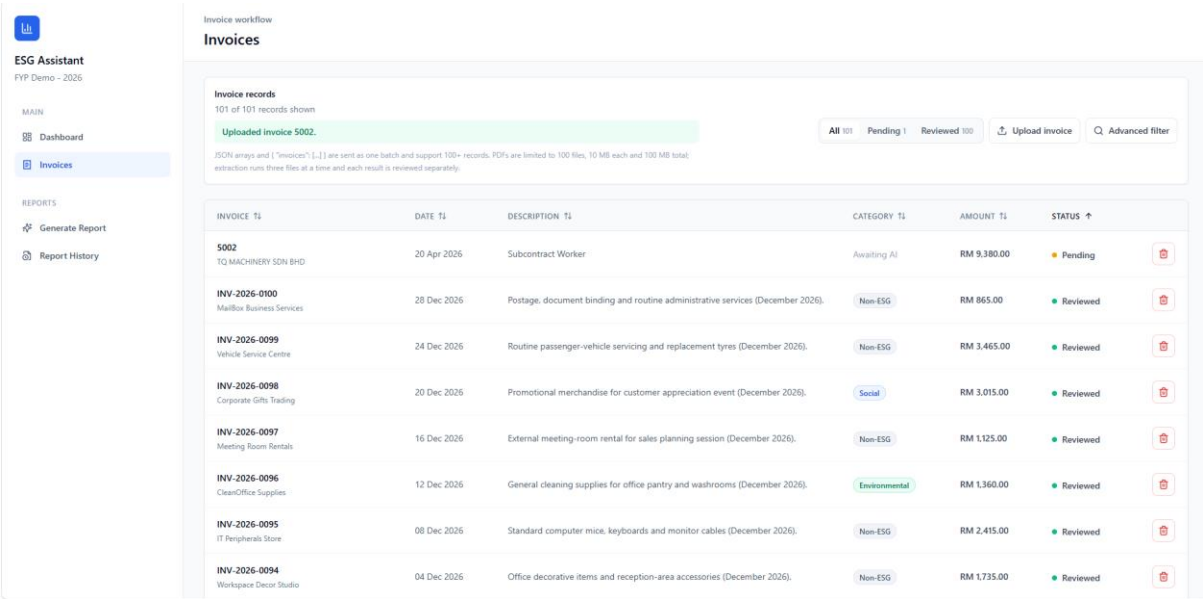


Figure 5.4.4: Newly imported invoice awaiting AI classification and human review.

After the background worker completed the classification task, the system displayed the AI prediction in the invoice-review panel. Figure 5.4.5 shows the predicted ESG category, AI-generated reason and evidence traceability information. The evidence traceability section presented the guideline source, ESG section, topic, page reference and supporting text retrieved through the local RAG pipeline. Users then reviewed the AI prediction and could either retain the predicted category or select a different final category. When the user selected **Save review**, the classification status changed from Pending to Reviewed.

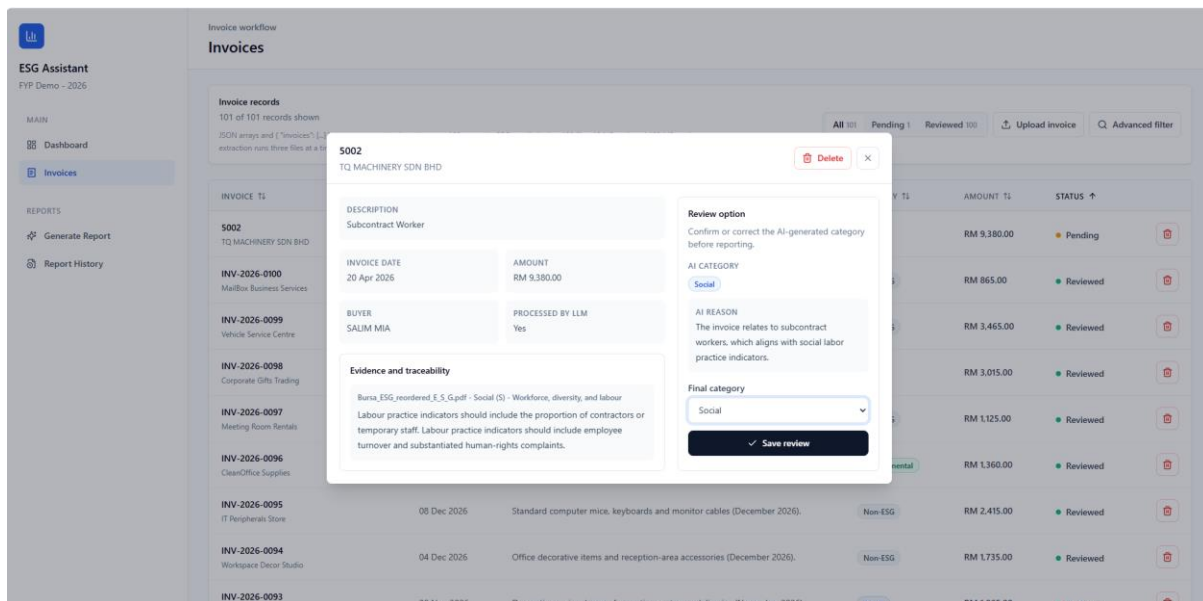


Figure 5.4.5: AI prediction, retrieved guideline evidence and human-review option.

This workflow combined automated ESG classification with human oversight. The AI prediction and retrieved guideline evidence supported the review decision, while the Reviewed status ensured that only human-confirmed invoice records were used for ESG report generation.

5.4.2 Agentic Report Generation and Export

The report-generation workflow was initiated from the Generate Report page. As shown in Figure 5.4.6, the user entered a report title and selected a reporting year. The system displayed the number of Reviewed invoices available for the selected period, as only human-confirmed invoice classifications were included in report generation. After the user selected **Generate draft report**, the system created a background report-agent run without blocking access to other pages.

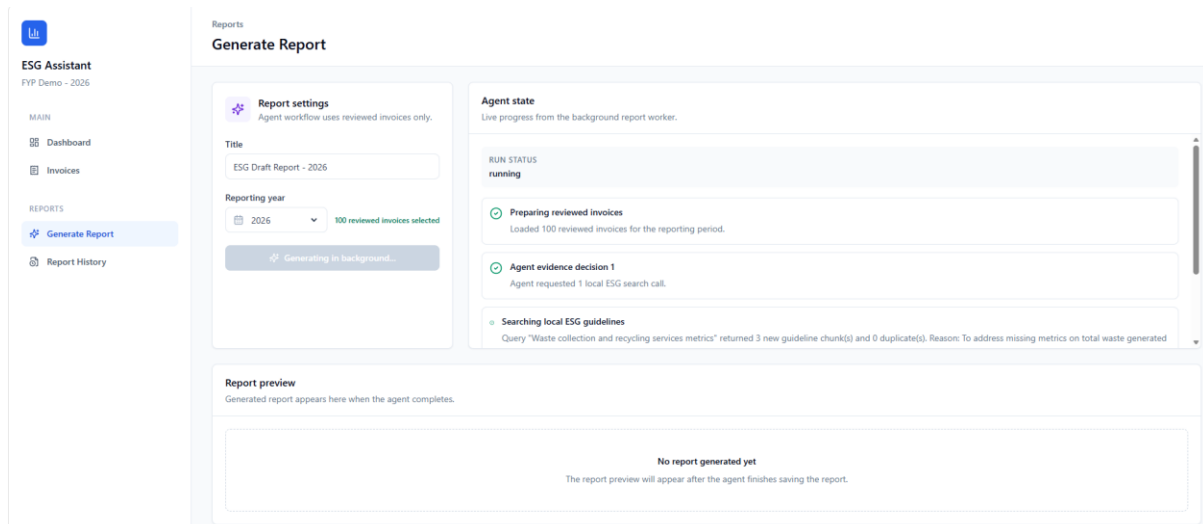


Figure 5.4.6: Generate Report page and live Agent State window.

The Agent State window displayed the progress of the background report-generation workflow. The frontend repeatedly retrieved the persisted run state and presented each completed or currently running step. These steps included preparing the reviewed invoice snapshot, making an agent evidence decision and searching the local ESG guideline database. Each guideline-search step displayed the query, the number of retrieved guideline chunks and the reason for the retrieval. This enabled users to monitor the agent's activities while report generation continued in the background.

After the agent completed the workflow and the report passed validation, the generated content was displayed in the Report Preview area below the Agent State window. Figure 5.4.7 shows an example of the completed report preview. The page also provided a **Download report** button, which exported the report package as a ZIP file.

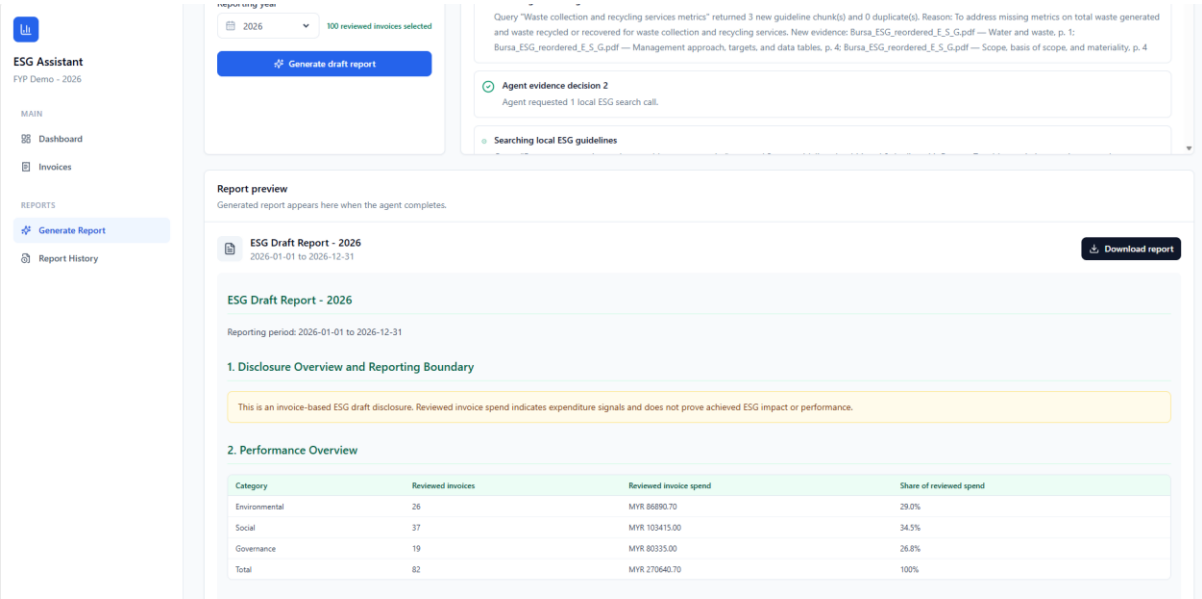


Figure 5.4.7: Generated ESG draft report displayed in the Report Preview area.

Previously generated reports were retained on the Report History page. As shown in Figure 5.4.8, each saved report record displayed its title, reporting period, version and draft status. Users could select **Preview** to inspect a saved report or select **Download report** to obtain the corresponding ZIP package. Each ZIP package contained the report in PDF format and its related CSV reference register.

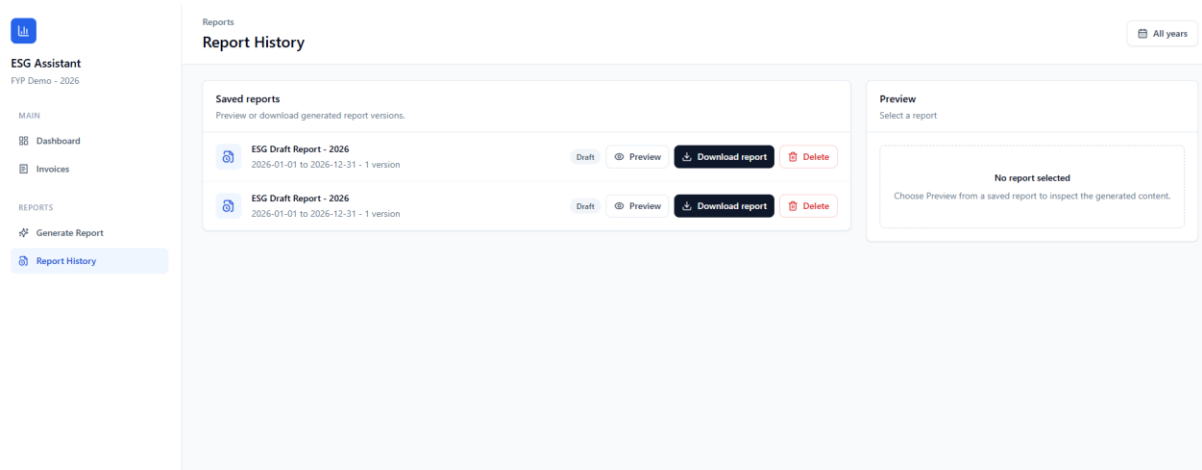


Figure 5.4.8: Report History page for previewing and downloading saved reports.

The generated report began with a disclosure overview and reporting boundary. This section clarified that the report was an invoice-based ESG draft disclosure and that reviewed invoice expenditure represented expenditure signals rather than verified ESG outcomes or impacts. The second section, Performance Overview, summarised the reviewed invoice records by ESG

category. As illustrated in Figure 5.4.9, the table presented the number of reviewed invoices, reviewed invoice expenditure and the share of reviewed expenditure for Environmental, Social and Governance categories, followed by the overall total.

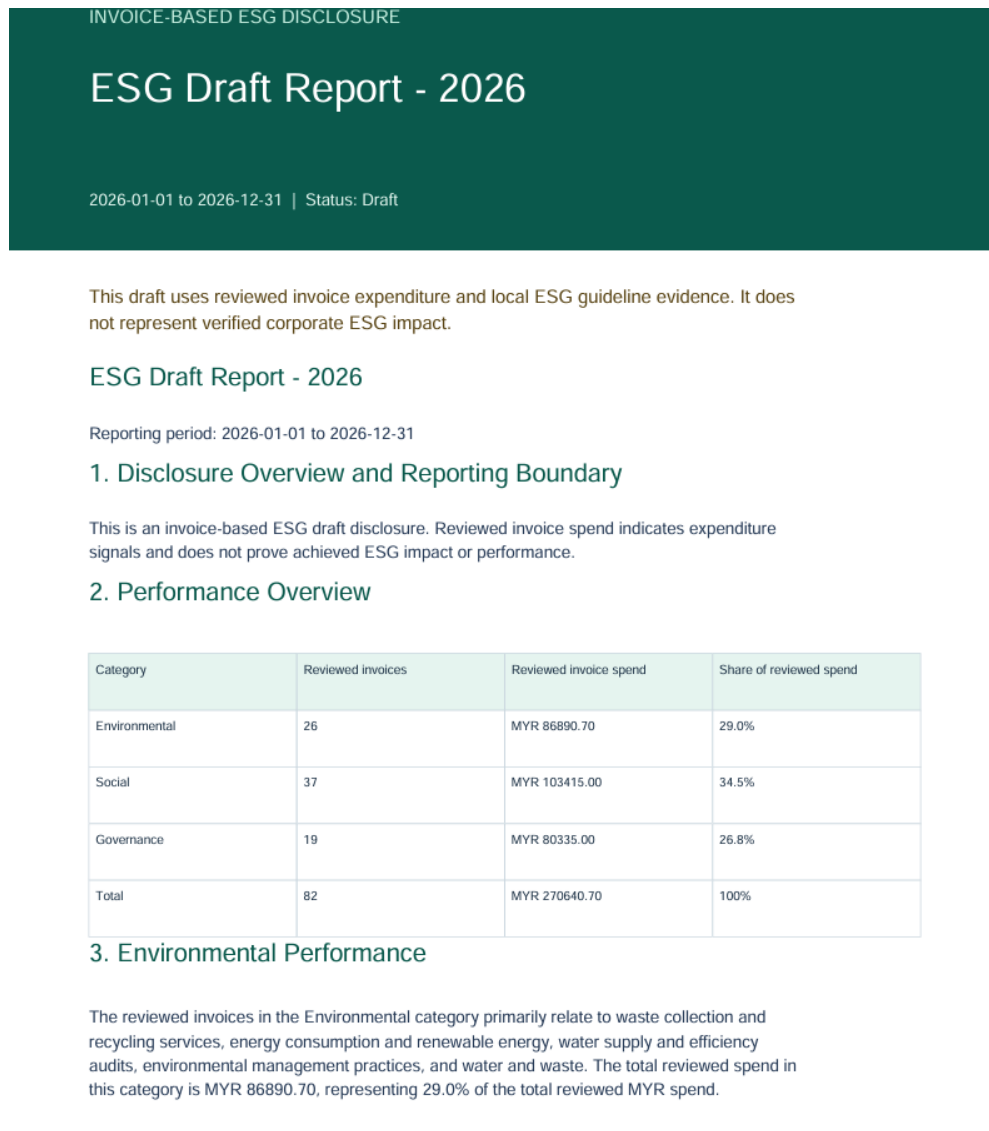


Figure 5.4.9: Disclosure overview and ESG performance overview in the generated report.

After the overall summary, the report presented separate Environmental, Social and Governance performance sections. Each section summarised the reviewed expenditure associated with the relevant ESG category and identified observed themes derived from the reviewed invoice descriptions. Figure 5.4.10 presents an Environmental example, including themes such as waste collection and recycling services, energy consumption and renewable energy, water supply and efficiency audits, and environmental management practices. Each theme included the total reviewed spend, an invoice-group reference such as [INV-GRP:E-01],

and guideline evidence containing the relevant source, topic, section and page reference. This structure connected each narrative statement to both the applicable ESG guideline and the supporting group of invoices.

Observed Themes
- Waste collection and recycling services
- Total reviewed spend: MYR 3795.00 · Reference: [INV-GRP:E-01]
- Evidence: (Source: Bursa_ESG_reordered_E_S_G.pdf; Topic: Water and waste; Section: Environmental (E); Page: 1)
- Energy consumption and renewable energy
- Total reviewed spend: MYR 45705.50 · Reference: [INV-GRP:E-02]
- Evidence: (Source: Bursa_ESG_reordered_E_S_G.pdf; Topic: Core environmental disclosures; Section: Environmental (E); Page: 1)
- Water supply and efficiency audits
- Total reviewed spend: MYR 10565.20 · Reference: [INV-GRP:E-03]
- Evidence: (Source: Bursa_ESG_reordered_E_S_G.pdf; Topic: Water and waste; Section: Environmental (E); Page: 1)
- Environmental management practices
- Total reviewed spend: MYR 25465.00 · Reference: [INV-GRP:E-04]
- Evidence: (Source: Bursa_ESG_reordered_E_S_G.pdf; Topic: Core environmental disclosures; Section: Environmental (E); Page: 1)
- Water and waste
- Total reviewed spend: MYR 1360.00 · Reference: [INV-GRP:E-05]
- Evidence: (Source: Bursa_ESG_reordered_E_S_G.pdf; Topic: Water and waste; Section: Environmental (E); Page: 1)

Figure 5.4.10: Example of observed themes and traceable invoice-group evidence in the Environmental performance section.

The CSV reference register acted as an appendix to the generated report. Figure 5.4.11 shows that each row linked an invoice to the invoice group cited in the report. The CSV file contained the Invoice Group, Invoice Number, Invoice Description, Amount, Date and Classification fields. Consequently, a reader could use an invoice-group reference in the report to identify the individual invoices that supported the related ESG theme.

Invoice Group	Invoice Number	Invoice Description	Amount	Date	Classification
[INV-GRP:E-01] Waste collection and recycling services	INV-2026-0012	Scheduled waste treatment at	MYR 1320.00	1/10/2026	Environmental
[INV-GRP:E-01] Waste collection and recycling services	INV-2026-0023	Scheduled waste treatment at	MYR 1495.00	2/23/2026	Environmental
[INV-GRP:E-01] Waste collection and recycling services	INV-2026-0003	Scheduled waste collection a	MYR 980.00	4/6/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0011	Rooftop solar system mainter	MYR 4850.00	1/6/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0014	Electric forklift lease and chai	MYR 3950.00	1/18/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0015	LED lighting retrofit for energy	MYR 6280.00	1/22/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0017	Recycled-content packaging r	MYR 3475.00	1/30/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0020	Renewable electricity certifica	MYR 3650.00	2/11/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0022	Rooftop solar system mainter	MYR 5025.00	2/19/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0025	Electric forklift lease and chai	MYR 4125.00	3/3/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0026	LED lighting retrofit for energy	MYR 6455.00	3/7/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0028	Recycled-content packaging r	MYR 3650.00	3/15/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0031	Renewable electricity certifica	MYR 3825.00	3/27/2026	Environmental
[INV-GRP:E-02] Energy consumption and renewable energy	INV-2026-0001	Monthly electricity bill for offi	MYR 420.50	4/6/2026	Environmental
[INV-GRP:E-03] Water supply and efficiency audits	INV-2026-0013	Water-efficiency audit and lea	MYR 2760.00	1/14/2026	Environmental

Figure 5.4.11: CSV reference register containing invoices associated with report invoice groups.

This workflow enabled users to generate, inspect, retain and export traceable ESG draft reports. The report content provided category-level ESG expenditure summaries, while the accompanying CSV register preserved the invoice-level evidence supporting the report narratives.

5.5 Implementation Issues and Challenges

5.5.1 Inconsistent Invoice PDF Formats

Invoice documents were received in different layouts and quality levels. Some PDF files contained selectable text and could be processed through direct text extraction, while others were scanned documents or image-based PDFs without sufficient embedded text. This could result in incomplete invoice numbers, supplier names, dates, descriptions or amounts.

To address this issue, the system first attempted direct PDF-text extraction. When extractable text was unavailable or insufficient, the system used Tesseract OCR as a fallback mechanism. The extracted information was then presented in an extraction-confirmation window before the invoice was saved. Users could review and amend the extracted invoice fields manually. This approach reduced the risk of incorrect source information being sent to the classification workflow.

5.5.2 Offline Embedding Model Availability

The initial embedding-model configuration depended on Hugging Face model loading. This caused a deployment issue when the required model files were not already cached and the application could not access the Hugging Face service. As a result, guideline indexing and retrieval could fail even though the ESG guideline PDF was available locally.

The issue was resolved by bundling the Sentence-Transformers MiniLM embedding model with the backend and loading it in local-only mode. The same local model was used for both guideline indexing and query embedding. This removed the dependency on model downloading during normal operation and enabled the Chroma vector database to support local ESG guideline retrieval without requiring an internet connection.

5.5.3 Reliability of AI-Generated Citations

An LLM could potentially generate invoice references, guideline references or ESG claims that were not supported by the available evidence. This was particularly important because the generated report was intended to provide traceable invoice-based ESG disclosure content.

Several validation mechanisms were implemented to reduce this risk. Report generation used a fixed snapshot of reviewed invoices from the selected reporting period. Invoice-group references were validated against this snapshot before the report was saved. Guideline citations were also validated against the exact Chroma chunk IDs that had been retrieved by the agent or were already attached to the selected invoice evidence. The system then retrieved the cited guideline chunk directly and rendered the public source, section, topic and page metadata automatically. This prevented unsupported or manually invented citations from being included in the final report.

5.5.4 Long-Running Classification and Report-Generation Tasks

Invoice classification and report generation could take longer than a normal web request because they involved extraction, RAG retrieval and LLM processing. If these tasks were

executed only within the active browser page, users could not comfortably continue using the system, and an interrupted session could cause the task status to be lost.

The system addressed this issue through persistent background processing. Invoice-classification tasks were stored as SQLite jobs and processed by background workers. Each job retained its status, retry attempts, worker lease and error information. Report generation was executed as a background LangGraph workflow, while the report-run status and individual agent steps were persisted in SQLite. The frontend repeatedly retrieved the latest run state and retained the active report-run ID in browser local storage. Therefore, users could navigate to other pages and return to the Generate Report page without interrupting the running task.

5.5.5 Readability of Report Evidence

Displaying every supporting invoice and guideline citation directly within the main report would have made the ESG report overly long and difficult to read. However, removing detailed evidence would have weakened invoice traceability.

This issue was addressed through a layered evidence design. The report grouped related invoices into thematic invoice groups, such as waste collection and recycling services or health, safety and employee wellbeing. Each observed theme displayed its total reviewed spend, an invoice-group reference and the related guideline evidence. The detailed invoice records were exported separately in the CSV reference register contained in the report ZIP package. This allowed the main report to remain readable while preserving invoice-level traceability in an accompanying appendix.

5.5.6 Controlling the Report-Agent Retrieval Process

An unrestricted report agent could continue requesting guideline searches unnecessarily, increasing execution time and API usage. Conversely, a fixed set of searches requested in advance would not allow the agent to evaluate evidence progressively.

The report workflow was configured as a decision → one retrieval → decision cycle. After each retrieval result, the agent evaluated the newly available guideline evidence before

deciding whether another focused search was necessary or whether it had sufficient evidence to produce the report. A maximum of eight local ESG guideline searches was applied to provide an execution boundary while maintaining agent-driven retrieval decisions.

Overall, these solutions focused on improving operational reliability, evidence traceability and usability during implementation. The effectiveness of the completed system was evaluated separately in Chapter 6.

5.6 Concluding Remark

This chapter described the implementation of the proposed ESG Assistant system. The system was developed as a web-based prototype using a Next.js frontend, FastAPI backend, SQLite database, Chroma vector database and locally bundled embedding model. The implementation included invoice import and text extraction, OCR fallback support, background AI classification, human review, ESG guideline retrieval and dashboard summaries.

The chapter also explained how Bursa Malaysia ESG guideline content was indexed into the local vector database to provide traceable RAG evidence. Background workers were used to process invoice-classification jobs, while a LangGraph-based report agent generated ESG draft reports from reviewed invoice records. The report-generation workflow preserved invoice and guideline evidence, displayed the agent's progress to users and exported the completed report together with a CSV reference register in a ZIP package.

The implementation challenges and the corresponding solutions were also presented, including inconsistent invoice PDF formats, offline embedding-model loading, citation validation, long-running background tasks and report-evidence readability. The following chapter evaluates the completed system based on its functional behaviour and the project evaluation criteria.

Chapter 6

SYSTEM EVALUATION AND DISCUSSION

6.1 System Testing and Performance Metrics

6.1.1 Evaluation Scope and Strategy

The evaluation scope covered the main workflow of the proposed system, beginning with invoice creation or PDF upload and ending with ESG report generation and export. The automated tests focused on the correctness and reliability of the main backend modules. These included invoice and batch processing, ESG classification with RAG evidence retrieval, OCR extraction, report validation, citation validation, and report export.

A total of 18 focused automated tests were designed across four core modules. The automated tests used temporary databases and mocked language model responses where appropriate. This ensured that software behaviour could be tested repeatedly without consuming API credits or depending on external network availability.

The ESG classification evaluation used a fixed dataset of 90 synthetic invoices with predefined expected ESG labels. The dataset contained 22 Environmental invoices, 21 Social invoices, 24 Governance invoices, and 23 Non-ESG invoices. The expected labels were not provided to the system during classification. They were used only after the system had generated a prediction, allowing the predicted category to be compared with the expected category.

The OCR evaluation used 12 controlled invoice PDFs. The dataset consisted of four native text-based PDFs, four clean scanned PDFs, and four moderately degraded scanned PDFs. Each PDF contained predefined ground-truth values for invoice number, invoice date, supplier name, total amount, and invoice description.

Finally, six manual end-to-end test cases were designed to evaluate the complete user workflow. These cases covered native PDF upload, OCR fallback, ESG classification with evidence, manual classification review, batch processing, and ESG report generation with export.

Evaluation Component	Scope	Test Data or Test Cases
Core automated testing	Invoice/batch, classification/RAG, OCR, report/export modules	18 automated tests
ESG classification and RAG evaluation	ESG category prediction and evidence attachment	90 labelled synthetic invoices
OCR extraction evaluation	Extraction of five invoice fields from different PDF conditions	12 invoice PDFs
End-to-end system testing	Complete workflow from invoice upload to report export	6 manual test cases

Table 6.1.1 Evaluation scope and testing strategy

6.1.2 Testing Metrics

Different metrics were used because each evaluation component assessed a different aspect of system performance. Automated testing and end-to-end testing focused on functional correctness, while ESG classification evaluation focused on category prediction performance. OCR evaluation focused on the accuracy of extracted invoice fields and processing time.

For ESG classification, Accuracy, Precision, Recall, and F1-score were used. Accuracy measured the proportion of invoices that were assigned the correct category. Precision measured how many invoices predicted under a category were actually correct, while Recall measured how many invoices belonging to a category were successfully identified. The F1-score represented the balance between Precision and Recall.

Because the system classified four categories, Macro Precision, Macro Recall, and Macro F1-score were used. These metrics calculated the average result across Environmental, Social, Governance, and Non-ESG categories, ensuring that each category contributed equally to the final evaluation. A confusion matrix was also used to identify common misclassification patterns between categories.

Evidence coverage was included as a traceability metric. It measured the proportion of classification results that contained at least one selected ESG guideline evidence item. However, evidence coverage did not by itself prove that the selected evidence was relevant or correct.

CHAPTER 6 SYSTEM EVALUATION AND DISCUSSION

Evidence relevance and citation correctness were therefore discussed qualitatively through error analysis and report validation.

For OCR evaluation, field accuracy measured whether each extracted field matched its predefined ground-truth value. Document success rate measured the proportion of invoice PDFs for which all five required fields were extracted correctly. Average processing time was also measured to compare the processing efficiency of native PDF text extraction and OCR fallback.

For the manual end-to-end test cases, each test case was recorded using its precondition, test steps, expected result, actual result, and final status. A test case was considered successful when the actual system behaviour matched the expected result.

Evaluation Area	Metric	Description
Automated testing	Test pass rate	Percentage of automated tests that passed successfully
ESG classification	Accuracy	Percentage of correctly predicted invoice categories
ESG classification	Precision, Recall and F1-score	Performance for each individual ESG category
ESG classification	Macro Precision, Macro Recall and Macro F1-score	Average classification performance across all four categories
ESG classification	Confusion matrix	Distribution of actual and predicted categories
RAG traceability	Evidence coverage	Percentage of predictions containing at least one selected evidence item
OCR extraction	Field accuracy	Percentage of correctly extracted invoice fields
OCR extraction	Document success rate	Percentage of PDFs with all required fields extracted correctly
OCR extraction	Average processing time	Mean time required to process an invoice PDF
End-to-end testing	Test case status	Whether actual system behaviour matched the expected result

Table 6.1.2 Performance metrics used for system evaluation.

6.2 Testing Setup and Result

6.2.1 Test Environment and Evaluation Datasets

The system evaluation was conducted on a local laptop environment. The system consisted of a Next.js frontend, a FastAPI backend, a SQLite database, a Chroma vector database, LangChain-based RAG components, a LangGraph report-generation workflow, and a Tesseract OCR fallback process. The test environment was configured to reflect the intended local deployment environment of the proposed prototype.

The hardware configuration used during testing is presented in Table 6.2.1.

Item	Configuration
Laptop	HP Laptop 15s-eq3xxx
Operating System	Windows 11 Home Single Language, Version 23H2
OS Build	22631.6199
System Type	64-bit operating system, x64-based processor
Processor	AMD Ryzen 5 5625U with Radeon Graphics, 2.30 GHz
Memory	8.00 GB RAM, with 7.34 GB usable
Graphics	AMD Radeon(TM) Graphics, 496 MB

Table 6.2.1 Hardware and operating system configuration used for system evaluation.

The main software components used during testing are summarised in Table 6.2.2. The language model temperature was set to 0.0 to reduce output variation during classification. The RAG component used similarity-based retrieval with a maximum of three retrieved guideline passages for each query.

Component	Technology or Version
Programming Language	Python 3.13.5
Backend Framework	FastAPI 0.115.6
Frontend Framework	Next.js 14.2.5 with React 18.3.1
JavaScript Runtime	Node.js 22.20.0 with npm 10.9.3
Relational Database	SQLite
Vector Database	Chroma 1.5.9

Component	Technology or Version
RAG Framework	LangChain 1.3.18
Report Agent Framework	LangGraph 1.2.11
Embedding Model	Locally stored all-MiniLM-L6-v2 embedding model
Language Model	OpenAI GPT-4o-mini
PDF Text Extraction	PyMuPDF 1.26.4
OCR Fallback	PyTesseract 0.3.13 with Tesseract OCR
Retrieval Configuration	Similarity search, top-k = 3
Classification Temperature	0.0

Table 6.2.2 Software environment and main system configuration.

Four evaluation datasets and test groups were used. The automated test suite consisted of 18 focused tests covering the core system modules. The ESG classification evaluation used 90 synthetic invoices with predefined expected categories. These expected labels were not submitted to the system during classification and were only used after prediction for performance comparison.

The invoice PDF text extraction and OCR fallback evaluation used 12 controlled invoice PDFs. The PDF set contained native text-based invoices, clean scanned invoices, and moderately degraded scanned invoices. Each PDF had predefined reference values for invoice number, date, supplier name, amount, and description. In addition, six manual end-to-end test cases were prepared to assess the complete workflow from invoice upload to ESG report export.

Evaluation Component	Dataset or Test Cases	Composition
Core automated testing	18 tests	Invoice/batch processing, classification/RAG, OCR, report/export
ESG classification evaluation	90 invoices	Environmental: 22; Social: 21; Governance: 24; Non-ESG: 23
Classification difficulty	90 invoices	65 clear cases and 25 hard cases
OCR evaluation	12 PDF invoices	4 native PDFs, 4 clean scans, and 4 degraded scans
OCR reference fields	5 fields per PDF	Invoice number, date, supplier, amount, and description

Evaluation Component	Dataset or Test Cases	Composition
End-to-end evaluation	6 manual test cases	Upload, OCR, classification, review, batch processing, and report export

Table 6.2.3 Evaluation datasets and manual test cases.

The core automated tests were executed using temporary SQLite databases and mocked language model responses. This approach ensured that software functions could be tested repeatedly without external API dependency. In contrast, the ESG classification experiment was executed using the actual RAG and GPT-4o-mini classification workflow. A separate evaluation database was used to store the 90 invoice records, predictions, evidence items, and calculated metrics. The OCR experiment was performed locally through the implemented PDF text extraction and Tesseract OCR fallback workflow.

6.2.2 Core Automated Testing Results

Core automated testing was conducted to verify the correctness and reliability of the main backend modules before evaluating the system with the invoice classification and OCR datasets. The test suite was executed using Python unittest discovery. Temporary SQLite databases were used for database-dependent tests, while language model responses were mocked where appropriate. This approach enabled the software functions to be evaluated repeatedly without requiring external API calls.

The final core test suite consisted of 18 automated tests across four modules: invoice and batch processing, ESG classification and RAG, invoice PDF extraction and OCR fallback, and report generation/export. All 18 tests were completed successfully.

Module	Number of Tests	Main Functions Evaluated	Result
Invoice and Batch Pipeline	4	- Invoice idempotency - Reclassification after invoice updates - Classification-job retry limit - Valid and invalid batch import handling	4/4 Passed
ESG Classification and RAG	5	- Selected evidence storage - Safe handling of invalid LLM output	5/5 Passed

Module	Number of Tests	Main Functions Evaluated	Result
		- Query rewrite - Query fallback - Conversion of retrieved documents into evidence items	
PDF Extraction and OCR Fallback	3	- Invoice field parsing - Native PDF text extraction - OCR fallback activation for image-only PDFs	3/3 Passed
Report Generation and Export	6	- Reviewed invoice summary - Guideline citation validation - Invoice snapshot validation - Report structure validation - Report snapshot persistence - PDF export	6/6 Passed
Total	18		18/18 Passed

Table 6.2.4 Results of the core automated test suite.

The automated test suite was executed through the backend terminal. As shown in Figure 6.2.1, all 18 core tests passed successfully, with a total execution time of 0.238 seconds.

```

C:\Users\Junbi\Documents\FYP\backend> .\venv\Scripts\python.exe -m unittest discover -s tests -v
test_classification_saves_only_selected_evidence (test_core_classification_rag.CoreClassificationAndRagTests.test_classification_saves_only_selected_evidence) ... ok
test_invalid_llm_structure_returns_safe_gateway_error (test_core_classification_rag.CoreClassificationAndRagTests.test_invalid_llm_structure_returns_safe_gateway_error) ... ok
test_query_rewrite_failure_falls_back_to_original (test_core_classification_rag.CoreClassificationAndRagTests.test_query_rewrite_failure_falls_back_to_original) ... ok
test_query_rewrite_keeps_original_and_deduplicates (test_core_classification_rag.CoreClassificationAndRagTests.test_query_rewrite_keeps_original_and_deduplicates) ... ok
test_rag_maps_documents_to_evidence_contract (test_core_classification_rag.CoreClassificationAndRagTests.test_rag_maps_documents_to_evidence_contract) ... ok
test_batch_api_accepts_valid_item_and_reports_invalid_item (test_core_invoice_pipeline.CoreInvoicePipelineTests.test_batch_api_accepts_valid_item_and_reports_invalid_item) ... ok
test_invoice_create_is_idempotent_and_enqueues_once (test_core_invoice_pipeline.CoreInvoicePipelineTests.test_invoice_create_is_idempotent_and_enqueues_once) ... ok
test_invoice_update_increments_version_and_requires_reclassification (test_core_invoice_pipeline.CoreInvoicePipelineTests.test_invoice_update_increments_version_and_requires_reclassification) ... ok
test_worker_retries_three_times_then_marks_batch_failed (test_core_invoice_pipeline.CoreInvoicePipelineTests.test_worker_retries_three_times_then_marks_batch_failed) ... ok
test_native_pdf_uses_text_extraction (test_core_ocr.CoreOcrAndExtractionTests.test_native_pdf_uses_text_extraction) ... ok
test_parser_extracts_required_invoice_fields (test_core_ocr.CoreOcrAndExtractionTests.test_parser_extracts_required_invoice_fields) ... ok
test_scanned_pdf_falls_back_to_ocr (test_core_ocr.CoreOcrAndExtractionTests.test_scanned_pdf_falls_back_to_ocr) ... ok
test_generation_saves_the_same_reviewed_invoice_snapshot (test_core_report_pipeline.CoreReportPipelineTests.test_generation_saves_the_same_reviewed_invoice_snapshot) ... ok
test_guideline_marker_is_exactly_validated_and_rendered (test_core_report_pipeline.CoreReportPipelineTests.test_guideline_marker_is_exactly_validated_and_rendered) ... ok
test_invoice_citations_must_belong_to_frozen_snapshot (test_core_report_pipeline.CoreReportPipelineTests.test_invoice_citations_must_belong_to_frozen_snapshot) ... ok
test_pdf_export_contains_cover_sections_and_page_number (test_core_report_pipeline.CoreReportPipelineTests.test_pdf_export_contains_cover_sections_and_page_number) ... ok
test_report_contract_rejects_missing_required_disclosure (test_core_report_pipeline.CoreReportPipelineTests.test_report_contract_rejects_missing_required_disclosure) ... ok
test_reviewed_summary_contains_currency_share_and_traceability (test_core_report_pipeline.CoreReportPipelineTests.test_reviewed_summary_contains_currency_share_and_traceability) ... ok
. ok

-----
Ran 18 tests in 0.238s

OK

```

Figure 6.2.1: Terminal Output of the Core Automated Test Suite

The invoice and batch pipeline tests verified that duplicate invoice submissions did not create duplicate records or duplicate classification jobs. They also verified that a modified invoice

created a new invoice version and required reclassification. In addition, the background worker was tested to ensure that classification jobs were retried up to three times before being marked as failed.

The ESG classification and RAG tests verified that only the evidence chunks selected by the classification model were stored with the final classification result. The tests also verified that invalid structured LLM responses were converted into a controlled gateway error instead of causing uncontrolled system failure. Query rewriting and fallback behaviour were tested to ensure that the original invoice description remained available when query rewriting failed.

The PDF extraction tests verified both processing paths of the invoice extraction pipeline. Native PDFs with embedded text were processed through direct PDF text extraction, while image-only PDFs triggered the OCR fallback process. The tests also confirmed that the parser could extract the required invoice fields from the processed text.

Finally, report-generation tests verified that the generated report contained the required report structure, used only reviewed invoice snapshots, validated invoice and guideline citations, and supported PDF export. These tests were important because the report module was designed to provide traceability between generated ESG narratives, invoice records, and retrieved guideline evidence.

The automated test pass rate was 100%. However, this result indicated that the tested software functions behaved correctly under the designed test conditions. It did not indicate that the ESG classification model achieved 100% classification accuracy. Classification effectiveness was evaluated separately using the 90-invoice dataset in Section 6.2.3.

6.2.3 ESG Classification and Evidence Analysis Results

This experiment evaluated the ESG classification performance of the complete RAG and GPT-4o-mini workflow. A total of 90 synthetic invoice descriptions with manually labelled reference labels were processed using the actual system pipeline. For each invoice, the system retrieved ESG guideline chunks, generated one predicted classification label, and stored the selected evidence item when available. The predicted labels were then compared with the validated reference labels.

CHAPTER 6 SYSTEM EVALUATION AND DISCUSSION

The dataset contained four classes: Environmental, Social, Governance, and Non-ESG. As shown in Table 6.2.5, the system correctly classified 76 out of 90 invoices, achieving an overall accuracy of 84.44%. The macro precision, macro recall, and macro F1-score were 88.35%, 85.10%, and 84.62%, respectively. These results indicate useful classification performance across the four classes within the controlled evaluation setting.

Metric	Result
Total invoices	90
Correct predictions	76
Accuracy	84.44%
Macro Precision	88.35%
Macro Recall	85.10%
Macro F1-score	84.62%
Evidence coverage	81.11%

Table 6.2.5: Overall ESG Classification and Evidence Results

Table 6.2.6 shows the classification performance for each ESG category. Environmental invoices achieved perfect results in this evaluation dataset. Social invoices achieved full recall, meaning all actual Social invoices were identified. However, its precision was lower because several Governance and Non-ESG invoices were incorrectly predicted as Social.

Actual Class	Precision	Recall	F1-score	Support
Environmental	95.65%	100.00%	97.78%	22
Social	63.64%	100.00%	77.78%	21
Governance	100.00%	70.83%	82.93%	24
Non-ESG	94.12%	69.57%	80.00%	23

Table 6.2.6: Classification Performance by ESG Category

The confusion matrix in Table 6.2.7 further shows the error pattern. All Environmental and Social invoices were correctly classified. Most errors involved Governance and Non-ESG invoices being classified as Social. Specifically, six Governance invoices and six Non-ESG invoices were predicted as Social. This indicates that invoices containing broad workplace,

service, employee, or community-related terms may be interpreted as Social-related even when their intended category was Governance or Non-ESG.

Predicted \ Actual	Environmental	Social	Governance	Non-ESG
Environmental	22	0	0	0
Social	0	21	0	0
Governance	0	6	17	1
Non-ESG	1	6	0	16

Table 6.2.7: Confusion Matrix of ESG Classification Results

The system recorded at least one selected evidence item for 81.11% of the evaluated invoices. This indicates that the RAG workflow attached selected guideline evidence to most predictions. However, evidence coverage only measures whether evidence was attached; it does not independently measure the semantic relevance, correctness, or ranking quality of the retrieved chunks.

Overall, the results demonstrate that the proposed system achieved good classification performance on the controlled evaluation dataset. Nevertheless, the confusion between Social, Governance, and Non-ESG categories suggests that more diverse real-world invoice descriptions and additional guideline examples would be required to further improve category separation.

6.2.4 Invoice PDF Text Extraction and OCR Fallback Evaluation

This evaluation examined the system's ability to extract essential invoice information from different PDF conditions. The implemented workflow first attempted direct text extraction using PyMuPDF. When a PDF did not contain extractable text, the system automatically activated the Tesseract OCR fallback process.

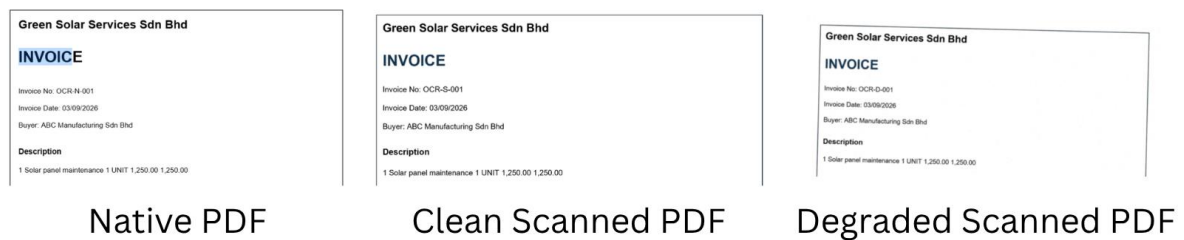


Figure 6.2.2 Representative Invoice PDFs Used for the Text Extraction and OCR Fallback Evaluation

Figure 6.2.2 presents one representative invoice from each PDF condition. Native text-based PDFs contain embedded text, whereas the clean and degraded scanned PDFs are image-based documents that require the OCR fallback workflow.

PDF Condition	No. of PDFs	Processing Method	Field Accuracy	Document Success Rate	Average Processing Time
Native text PDF	4	Direct PDF text extraction	100.00%	100.00%	0.01 s
Clean scanned PDF	4	Tesseract OCR fallback	100.00%	100.00%	0.83 s
Degraded scanned PDF	4	Tesseract OCR fallback	100.00%	100.00%	0.73 s
Overall	12	Hybrid extraction workflow	100.00%	100.00%	0.52 s

Table 6.2.8: PDF Text Extraction and OCR Fallback Evaluation Results

As shown in Table 6.2.8, all 12 test invoices were processed successfully, and all 60 required fields were extracted correctly. Native text-based PDFs achieved the lowest processing time because their embedded text could be extracted directly without OCR. In contrast, scanned PDFs required the OCR fallback process and therefore required a longer processing time.

The difference between the two scanned-PDF average processing times should not be interpreted as degraded scans being faster. OCR processing time can vary due to local machine workload, image preparation, and OCR engine warm-up. The main result is that the fallback workflow successfully extracted the required fields from both clean and moderately degraded scanned invoices in this controlled dataset.

Overall, the result confirms that the hybrid workflow supports both digitally generated invoices and image-based scanned invoices. However, the dataset was controlled and limited to 12 PDFs; performance may decrease for real invoices with severe blur, handwriting, complex tables, unusual layouts, or poor image resolution.

6.2.5 End-to-End System and Report Export Testing

End-to-end testing was conducted manually to verify that the main system functions worked together from invoice input to final report export. Unlike the automated unit tests, these test cases focused on the complete user workflow through the system interface, including PDF upload, information extraction, ESG classification, human review, batch processing, and report generation.

Six end-to-end test cases were prepared and executed through the system interface. The actual outcome and status of each case were recorded after testing.

Test Case 1: Native PDF Invoice Upload

Item	Details
Test No.	1
Test Case	Verify that the system can upload and extract information from a native text-based PDF invoice.
Description	This test verifies the direct PDF text extraction workflow for invoices containing embedded, selectable text.
Test Step	1. Navigate to the invoice upload page. 2. Select a native text-based invoice PDF. 3. Upload the PDF. 4. View the extracted invoice information.
Expected Result	The system should upload the invoice successfully and display the extracted invoice number, date, supplier name, amount, and description.
Actual Result	The system successfully uploaded the native text-based PDF invoice and identified the extraction method as PDF text. The invoice number, date, supplier, buyer, amount,

Item	Details
	currency, and description were displayed in the confirmation form for user review before import.
Status	Pass

Confirm extracted invoice
 Extraction method: PDF text
 Subcontract - 5005.pdf

Review needed
Please verify extracted fields before importing.

Invoice number
5005

Invoice date
04/20/2026

Supplier
TQ MACHINERY SDN BHD

Buyer
MUHAMAD AFIZI BIN AHAMAD

Amount
9438

Currency
MYR

Description
Subcontract Worker

Extracted text

tqmachinery55@gmail.com

CASH SALE / INVOICE

No
: 5005
MUHAMAD AFIZI BIN AHAMAD

Your P/O No
:

NO 220-B,

Order Date
:

BELAKANG NEW STORE,

Attention

Cancel **✓ Confirm import**

Figure 6.2.3: Successful Extraction of a Native Text-Based PDF Invoice

Test Case 2: Scanned PDF Invoice Upload

Item	Details
Test No.	2
Test Case	Verify that the system can extract information from an image-based scanned PDF invoice using OCR fallback.
Description	This test verifies that the system can process a scanned invoice that does not contain embedded text.
Test Step	1. Navigate to the invoice upload page. 2. Select an image-based scanned invoice PDF. 3. Upload the PDF. 4. View the extracted invoice information.

Item	Details
Expected Result	The system should activate the OCR fallback workflow and display the extracted invoice information correctly.
Actual Result	The system successfully detected that the uploaded PDF required OCR processing, as indicated by the “Extraction method: OCR” label. The invoice number, date, supplier, buyer, amount, currency, and description were extracted and displayed for review before import.
Status	Pass

Confirm extracted invoice

Extraction method: OCR

Screenshot_5.pdf

Review needed

Please verify extracted fields before importing.

Extracted text

Tw! TQ MACHINERY SDN BHD peo260:008705-1688803)
NO 69, LORONG BINTANG /NDAH 4, TAMAN BINTANG INDAH,
TQ MACHINERY 34900 PANTAI REMIS, PERAK
Tel: 010-3136669
tamachinery55@gmail.com
CASH SALE / INVOICE
No : 9002
SALIM MIA | Your P/O No :
NO 3-1, JALAN MEDAN TAMING 36, PUSAT Order Date : 7
PERNIAGAAN TAMIN JAYA 1
TAMAN TAMIN JAYS, Attention :
43300 SERJ KEMBANGAN, SELANGOR Date : 20/4/2026
PASSPORT NO: ELO544250 Page : of 1
TEL : 011-2600 0528
Item Description Qty UOM U/Price Disc.
Total
RM RM
1. Subcontract Worker 1 PC 9380.00
9380.00
REFER :
CUST PLC :
RINGGIT MALAYSIA
Notes :

Invoice number

5002

Supplier

TQ MACHINERY SDN BHD

Amount

9380

Description

Subcontract Worker

Invoice date

04/20/2026

Buyer

SALIM MIA |

Currency

MYR

Cancel

✓ Confirm import

Figure 6.2.4: OCR Fallback Extraction for an Image-Based Scanned Invoice

Test Case 6.3: ESG Classification with Guideline Evidence

Item	Details
Test No.	3
Test Case	Verify that the system can classify an invoice and provide ESG guideline evidence.

Item	Details
Description	This test verifies the integrated RAG and LLM classification workflow after invoice information has been extracted.
Test Step	1. Upload or select a processed invoice. 2. Submit the invoice for ESG classification. 3. Wait for the classification process to complete. 4. Open the classification result.
Expected Result	The system should assign one ESG category and display the classification reason and selected guideline evidence.
Actual Result	The system classified invoice 5003 as Social. It displayed an AI-generated reason relating the subcontract-worker expense to social labour practices and provided a relevant ESG guideline evidence item for traceability.
Status	Pass

5003

TQ MACHINERY SDN BHD

Delete

×

DESCRIPTION

Subcontract Worker

INVOICE DATE

20 Apr 2026

AMOUNT

RM 7,720.00

BUYER

MELVI AWAH

PROCESSED BY LLM

Yes

Evidence and traceability

Bursa_ESG_reordered_E_S_G.pdf - Social (S) - Workforce, diversity, and labour

Labour practice indicators should include the proportion of contractors or temporary staff. Labour practice indicators should include employee turnover and substantiated human-rights complaints.

Review option

Confirm or correct the AI-generated category before reporting.

AI CATEGORY

Social

AI REASON

The invoice relates to subcontract workers, which aligns with social labor practice indicators.

Final category

Social

Save review

Figure 6.2.5: AI-Generated ESG Classification with Guideline Evidence

Test Case 6.4: Manual Classification Review

CHAPTER 6 SYSTEM EVALUATION AND DISCUSSION

Item	Details
Test No.	4
Test Case	Verify that a user can review and update an invoice classification.
Description	This test verifies the human-review function that allows a user to confirm or revise an AI-generated classification.
Test Step	1. Open a classified invoice. 2. Review the predicted category and evidence. 3. Confirm the result or change the category. 4. Save the reviewed classification. 5. Refresh the invoice or dashboard page.
Expected Result	The reviewed classification should be saved successfully and reflected in the invoice record and dashboard summary.
Actual Result	The classification review for invoice 5003 was saved successfully. The invoice remained categorised as Social and its status was updated to “Reviewed” in the invoice list.
Status	Pass

INV-2026-0058 CyberGuard Assurance	13 Jul 2026	Cybersecurity controls audit and vulnerability assessment (July 2026).	Governance	RM 5,760.00	Reviewed	
INV-2026-0054 SecureAudit Consulting	27 Jun 2026	Internal controls audit and corporate-governance review service (June 2026).	Governance	RM 4,650.00	Reviewed	
5003 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Social	RM 7,720.00	Reviewed	
INV-2026-0028 CircularPack Industries	15 Mar 2026	Recycled-content packaging materials for finished goods (March 2026).	Environmental	RM 3,650.00	Reviewed	
INV-2026-0017 CircularPack Industries	30 Jan 2026	Recycled-content packaging materials for finished goods (January 2026).	Environmental	RM 3,475.00	Reviewed	

Figure 6.2.6: Reviewed Invoice Classification Displayed in the Invoice List

Test Case 6.5: Batch Invoice Processing

Item	Details
Test No.	5
Test Case	Verify that the system can process multiple invoices through batch upload.
Description	This test verifies the batch-processing workflow, including invoice submission, processing progress, and individual processing outcomes.
Test Step	1. Navigate to the batch upload page. 2. Select multiple invoice files. 3. Submit the batch. 4. Monitor the processing progress. 5. Open the processed invoice records.
Expected Result	The system should accept multiple invoice files, create a separate record for each invoice, and place the records in the classification queue with a pending status.
Actual Result	The system accepted multiple uploaded invoices and created separate invoice records for each file. The records were displayed in the invoice list with the status “Pending” and category “Awaiting AI”, indicating that they had entered the classification queue.
Status	Pass

INVOICE T1	DATE T1	DESCRIPTION T1	CATEGORY T1	AMOUNT T1	STATUS ↑	
5008 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Awaiting AI	RM 5,121.25	Pending	
5007 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Awaiting AI	RM 5,080.00	Pending	
5006 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Awaiting AI	RM 6,960.00	Pending	
5005 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Awaiting AI	RM 9,438.00	Pending	
5004 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Awaiting AI	RM 7,820.00	Pending	
5002 TQ MACHINERY SDN BHD	20 Apr 2026	Subcontract Worker	Social	RM 9,380.00	Pending	

Figure 6.2.7: Batch-Uploaded Invoices Listed as Pending Classification Jobs

Test Case 6.6: Report Generation and Export

CHAPTER 6 SYSTEM EVALUATION AND DISCUSSION

Item	Details
Test No.	6
Test Case	Verify that the system can generate and export an ESG report from reviewed invoice records.
Description	This test verifies the final reporting workflow, including report generation and export of the completed output.
Test Step	1. Ensure that reviewed invoice records are available. 2. Navigate to the report generation page. 3. Select the required reporting period or invoice records. 4. Generate the report. 5. Export the generated report.
Expected Result	The system should generate a report using the reviewed invoice data and provide the selected export file successfully.
Actual Result	The system successfully generated an ESG Draft Report for the selected reporting period. The report was available for preview, and the export package was downloaded successfully as a ZIP file.
Status	Pass

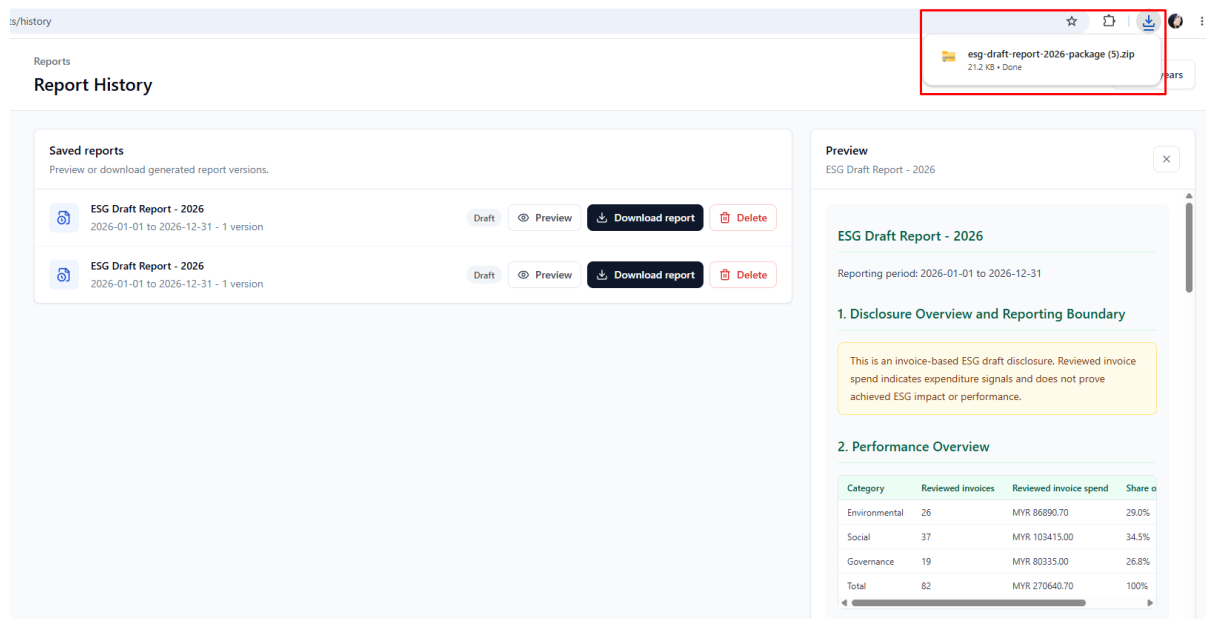


Figure 6.2.8: Generated ESG Report Preview and Downloaded Export Package

6.3 Project Challenges

6.3.1 Dataset and Ground-Truth Limitations

A major challenge was the limited availability of suitable invoice datasets with verified ESG labels. Real organisational invoices may contain confidential supplier, buyer, financial, and operational information. Therefore, publicly available invoice data that can be used for ESG classification evaluation is limited.

To address this limitation, a controlled dataset of 90 synthetic invoice descriptions with manually labelled reference labels was prepared for the ESG classification experiment. The dataset included Environmental, Social, Governance, and Non-ESG examples, as well as both clear and more ambiguous invoice descriptions. However, this dataset cannot fully represent the diversity of real-world invoices. Actual invoices may use inconsistent descriptions, abbreviations, incomplete information, multiple languages, or expense items that relate to more than one ESG category.

The manually validated reference labels were used for evaluation. Although this enabled the calculation of accuracy, precision, recall, and F1-score, ESG classification can involve

subjective interpretation in ambiguous cases. For example, certain workforce-related services may be interpreted as Social expenses, whereas internal control or compliance services may be more appropriately classified as Governance expenses. A larger dataset reviewed by multiple ESG domain experts would further improve the reliability of the reference labels in future work.

6.3.2 RAG, LLM and OCR Limitations

The RAG and LLM workflow depends on the content and coverage of the ESG guideline documents stored in the vector database. If a guideline does not contain sufficiently specific information for an invoice description, the model may retrieve less relevant evidence or produce an uncertain classification. This was reflected in the classification results, where some Governance and Non-ESG invoices were predicted as Social due to overlapping terms related to employees, services, or workplace activities.

The LLM-based classification process also depends on an external API. This introduces API cost, network dependency, response-time variation, and the possibility of slightly different outputs for similar inputs. The system reduces these risks through structured output validation, evidence storage, and a manual review function. Nevertheless, human review remains important before the generated classification is used in a final report.

For the OCR workflow, the controlled PDF evaluation achieved full extraction accuracy. However, the tested invoices had relatively consistent layouts and readable text. In real deployment, OCR performance may decrease for severely blurred scans, handwritten invoices, unusual table structures, skewed images, missing fields, or low-resolution documents. The current system therefore presents extracted fields for user confirmation before importing the invoice.

Overall, these challenges show that the system should be treated as a decision-support and reporting-assistance tool rather than a fully autonomous ESG reporting solution.

6.4 Evaluation Against Project Aims

This section evaluates the implemented prototype against the main project aims described in Chapter 1. The evaluation considers the development of the ESG knowledge base, the ESG classification workflow, and the end-to-end ESG reporting assistant.

Project Aim	Evaluation Evidence and Result	Conclusion and Limitation
Development of an ESG knowledge base	ESG guideline content was processed into retrievable chunks and vector embeddings, then integrated into the application workflow.	The knowledge base was successfully implemented and provided contextual evidence. Retrieval relevance was not independently assessed by domain experts.
Development of an ESG classification workflow	Using 90 manually validated synthetic invoice descriptions, the system achieved 84.44% accuracy, 88.35% macro precision, 85.10% macro recall, and 84.62% macro F1-score.	The workflow achieved useful performance in the controlled evaluation. Results may not generalise to all real-world invoice descriptions.
Development of an end-to-end ESG reporting assistant	All 18 automated tests passed. OCR achieved 100% field accuracy across 60 fields from 12 controlled PDF invoices, and all six end-to-end cases passed.	The prototype demonstrated end-to-end feasibility under controlled conditions. Further testing with diverse real invoices is required.

Table 6.4.1 Evaluation Against Project Aims

Overall, the evaluation indicated that the implemented prototype achieved the main project aims within a controlled environment. The system successfully indexed ESG guideline content, classified manually validated synthetic invoice descriptions, extracted invoice fields from controlled PDF documents, and generated outputs with retrieved supporting evidence. Nevertheless, the findings represent prototype-level results. Further validation using representative real-world invoices, independent retrieval-relevance assessment, and broader user or domain-expert evaluation is required before operational deployment can be considered.

6.5 Concluding Remark

This chapter presented the testing and evaluation results of the ESG Reporting Assistant. The results demonstrated that the core system components, including ESG knowledge retrieval, invoice classification, OCR extraction, and end-to-end workflow processing, operated successfully in the controlled evaluation environment. The next chapter concludes the project, summarises its contributions, and presents recommendations for future improvement.

Chapter 7

CONCLUSION AND RECOMMENDATION

7.1 Conclusion

This project designed, implemented, and evaluated an ESG Reporting Assistant that combined OCR, retrieval-augmented generation, and large language model-based classification for e-invoice analysis. The prototype was developed to support the identification of Environmental, Social, Governance, and Non-ESG information from invoice descriptions while presenting retrieved ESG guideline content as supporting context.

First, the project successfully developed a local ESG knowledge base by processing ESG reporting guideline content into retrievable text chunks and vector embeddings. The knowledge base was integrated into the application workflow and enabled the system to retrieve contextual ESG guidance when processing invoice descriptions.

Second, the implemented classification workflow was evaluated using 90 synthetic invoice descriptions with manually labelled reference labels. The system achieved an accuracy of 84.44%, a macro precision of 88.35%, a macro recall of 85.10%, and a macro F1-score of 84.62%. These findings indicate useful classification performance within the controlled evaluation setting across the Environmental, Social, Governance, and Non-ESG categories.

Third, the prototype demonstrated end-to-end feasibility through automated testing, OCR evaluation, and workflow testing. All 18 automated tests passed. The OCR evaluation achieved 100% field accuracy across 60 expected fields from 12 controlled PDF invoices, while all six end-to-end workflow cases passed. These results showed that the system could extract invoice information, perform ESG classification, retrieve supporting content, and present traceable outputs within the defined test environment.

Despite these positive results, the evaluation was conducted using synthetic invoice descriptions and controlled PDF files. Therefore, the findings should be interpreted as evidence of prototype feasibility rather than production-scale performance. Future validation should use

representative real-world invoices, independently assess retrieval relevance and evidence faithfulness, and involve intended users or ESG domain experts to evaluate practical usability and decision-support value.

7.2 Recommendation

Several improvements can be considered in future work to enhance the reliability, scalability, and practical value of the system.

First, the evaluation dataset can be expanded using more diverse real-world invoices, subject to appropriate privacy protection. A larger dataset containing invoices from different industries, suppliers, languages, layouts, and ESG scenarios would provide a more reliable measurement of the system's real-world classification performance. In addition, reference labels should ideally be reviewed by multiple ESG or accounting domain experts to reduce subjectivity in ambiguous cases.

Second, the classification workflow can be improved by introducing confidence scores and clearer handling for uncertain cases. For example, when the model has low confidence or retrieves weak guideline evidence, the system could automatically mark the invoice as "Review Required" rather than assigning a final ESG category immediately. A multi-label or hierarchical classification approach may also be considered for invoices that relate to more than one ESG dimension.

Third, the RAG component can be enhanced by incorporating additional ESG standards, sector-specific guidelines, and more detailed metadata. Future work may also evaluate retrieval quality separately through metrics such as evidence relevance and expert assessment of retrieved guideline chunks. This would provide deeper insight into whether the system retrieves the most appropriate evidence before classification.

Fourth, the OCR workflow can be improved through image pre-processing techniques, such as deskewing, noise reduction, contrast enhancement, and layout detection. These improvements may increase extraction reliability for low-resolution, skewed, handwritten, or highly degraded invoice documents.

CHAPTER 7 CONCLUSION AND RECOMMENDATION

Finally, future deployment should consider stronger production features, including user authentication, role-based access control, audit logs, secure data backup, configurable report templates, and monitoring of API usage. These features would help the system become more suitable for organisational use where invoice data and ESG reports may contain sensitive business information.

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POSTER

Design and Implementation of an ESG Reporting Assistant Using E-Invoice Data and Large Language Model

FACULTY OF INFORMATION COMMUNICATION AND TECHNOLOGY

CONTEXT SUMMARY

Current ESG reporting relies on manual, subjective judgment to analyze dispersed data. This project proposes an **automated assistant** that uses **e-invoices** as standardized digital records, processed through **Retrieval-Augmented Generation (RAG)** and **Large Language Models (LLMs)** to ensure efficient and evidence-based reporting.



PROJECT FOUNDATION & OBJECTIVES



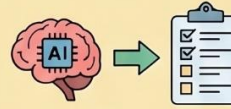
THE MANUAL REPORTING CHALLENGE

ESG reporting faces dispersed data, subjective judgment, and high manual processing costs.



E-INVOICES AS DATA SOURCES

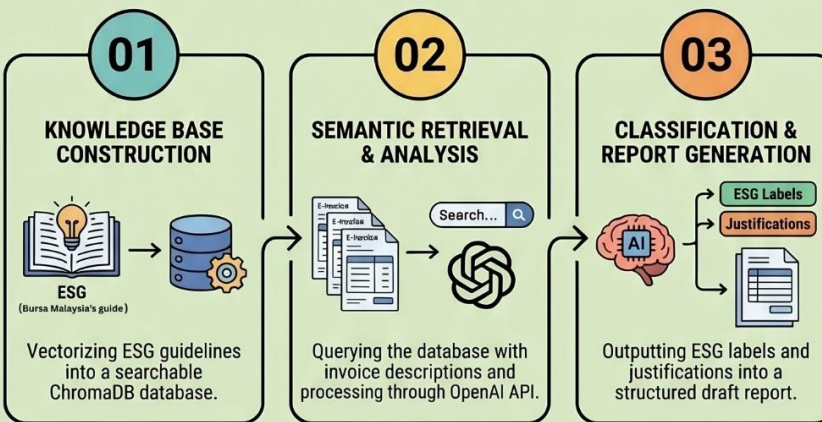
Standardized digital records serve as the primary analysis input.



PROJECT OBJECTIVE

Develop a RAG-integrated LLM prototype to improve efficiency and traceability in ESG assessments.

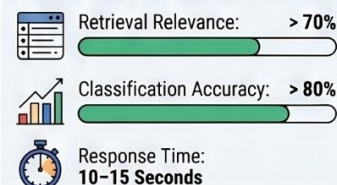
THE 3-STEP IMPLEMENTATION METHOD



WHY THIS SYSTEM IS SUPERIOR

- ENHANCED EFFICIENCY & CONSISTENCY:** Automates manual screening to reduce time and eliminate subjective human errors.
- TRACEABILITY & EVIDENCE-BASED RESULTS:** Every classification is linked to specific guidelines, reducing AI "hallucinations."
- BRIDGING THE DATA-STANDARD GAP:** Successfully maps raw transactional descriptions into formal ESG disclosure frameworks.

PERFORMANCE TARGETS



PROJECT DEVELOPER: LAM JUN BIN | PROJECT SUPERVISOR: PROF. TS. LIEW SOUNG YUE